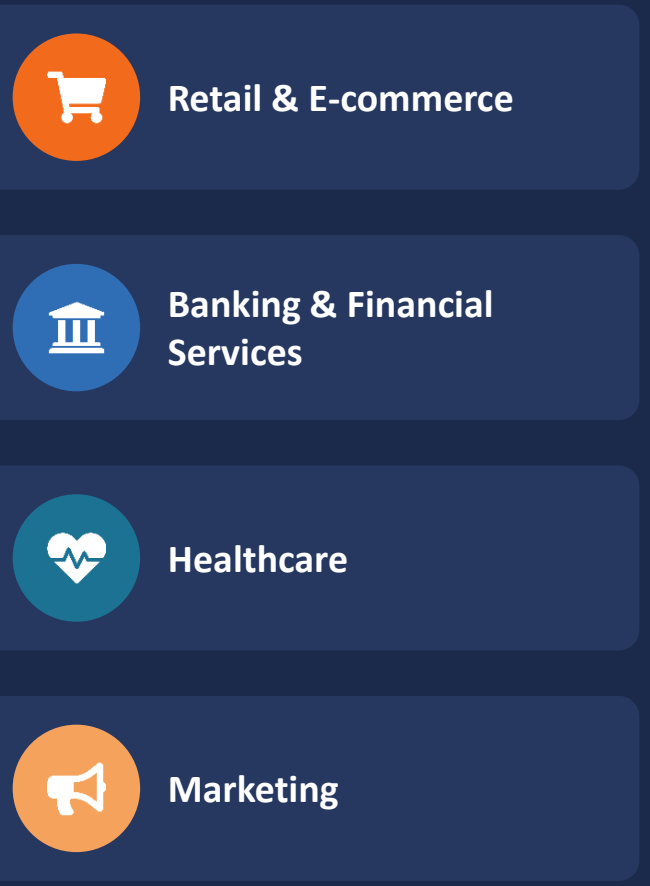
PORTFOLIO PROJECT

Multi-Domain Business Analytics Using SQL Server and Power BI

SQL Analysis, Business Intelligence and Data-Driven Insights

Author: Aman Ade

Tools: SQL Server | Power BI | CSV | GitHub



Executive Summary

Purpose

Demonstrate an end-to-end analytics workflow: data stored in Microsoft SQL Server, analyzed with SQL, exported to CSV and visualized in Power BI.

Scope

Four business domains (Retail and E-commerce, BFSI, Healthcare, Marketing) analyzed through 15 business questions.

Status of findings

Insight statements in this deck are provisional examples. They must be validated against the actual SQL outputs and Power BI pages.

4
Business domains

15
Business questions

15
Power BI report pages

2–3
Charts per question (approx.)

Business Problem



Sales performance

Which periods, products and regions drive sales and profit?



Customer value

How do customers differ in behavior and value?



Financial risk

Where do lending approvals and defaults need monitoring?



Healthcare operations

How are visits, departments and utilization changing?



Marketing performance

Which campaigns and channels return the most?

Common thread

Structured data analysis turns raw records into comparable metrics that support decisions.

Project Objectives

1

Database analysis

Explore the dataset stored in SQL Server.

2

Relational data handling

Work with related tables, keys and date fields.

3

SQL query development

Write queries that answer 15 business questions.

4

CSV export

Save query outputs to the SQL_Outputs folder.

5

Dashboard creation

Build a 15-page Power BI report.

6

Insight communication

Present findings clearly to business audiences.

Tools and Technologies



SQL Server

Relational database that stores the dataset and runs the analysis queries.



SQL

Query language used to retrieve and summarize data for each business question.



Power BI

Builds the 15-page report with approximately 2–3 charts per question.



CSV / Excel

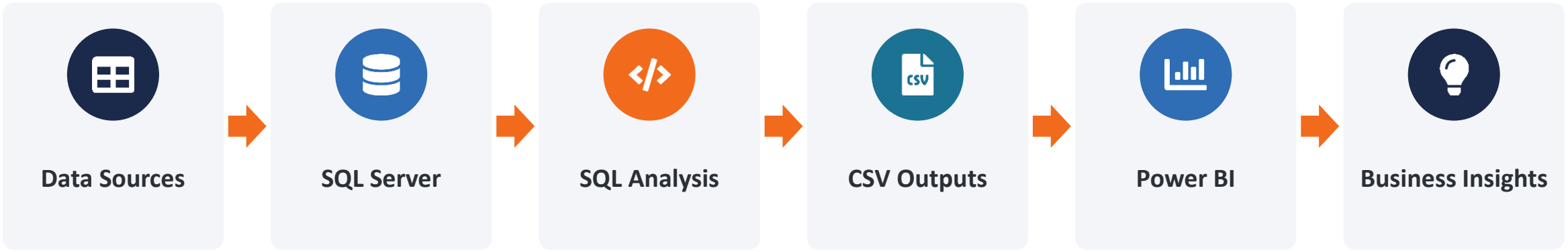
Query outputs saved as CSV files in the SQL_Outputs folder.



GitHub

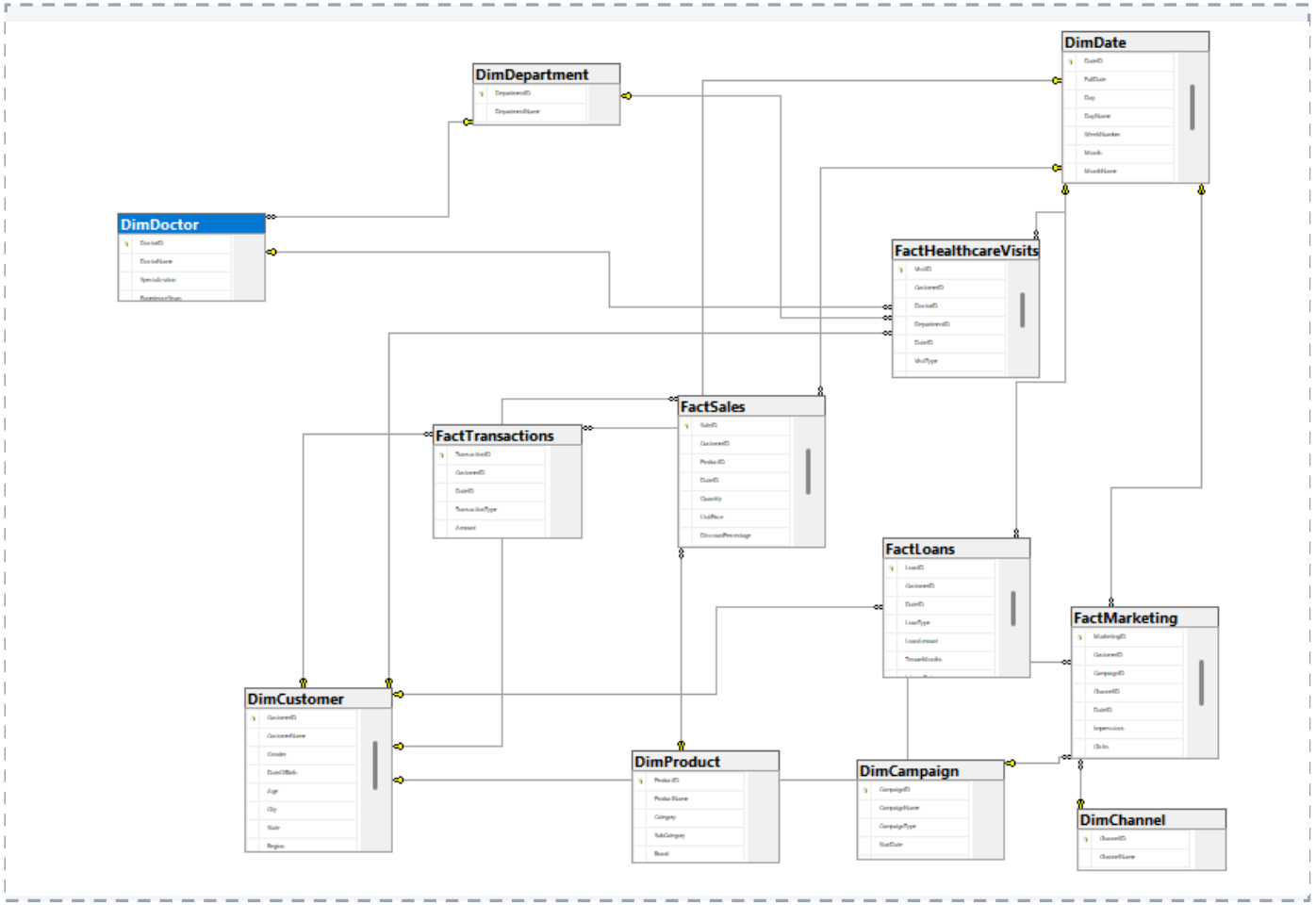
Version control and portfolio hosting for scripts, outputs and documentation.

End-to-End Workflow



Dataset → SQL Server Database → SQL Queries → SQL Query Outputs → CSV Exports → Power BI Visualizations → Business Insights

Database Structure



GENERAL CONCEPTS ONLY

Fact tables
Store measurable events such as transactions, sales or visits.

Dimension tables
Describe entities such as customers, products or departments.

Primary keys
Uniquely identify each row in a table.

Foreign keys
Link rows to related rows in another table.

Date relationships
Date fields enable time-based analysis.

Q1. What is the monthly sales and profit trend?

BUSINESS QUESTION

What is the monthly sales and profit trend?

BUSINESS OBJECTIVE

Understand how sales and profit change from month to month and where profitability shifts.

SQL QUERY

```
SELECT
  d.Year,
  d.Month,
  d.MonthName,
  SUM(f.SalesAmount) AS TotalSales,
  SUM(f.CostAmount) AS TotalCost,
  SUM(f.ProfitAmount) AS TotalProfit,
  CAST(
    100.0 * SUM(f.ProfitAmount)
    / NULLIF(SUM(f.SalesAmount), 0)
    AS DECIMAL(10,2)
  ) AS ProfitMarginPct
FROM dbo.FactSales f
JOIN dbo.DimDate d
  ON f.DateID = d.DateID
GROUP BY
  d.Year,
  d.Month,
  d.MonthName;
```

	Year	Month	MonthName	TotalSales	TotalCost	TotalProfit	ProfitMarginPct
1	2024	2	February	262831.56	238269.82	24561.74	9.35
2	2024	3	March	867082.63	810255.11	56827.52	6.55
3	2024	4	April	530625.63	529473.48	1152.15	0.22
4	2024	5	May	917381.07	840926.38	76454.69	8.33
5	2024	6	June	2267847.03	2094475.02	173372.01	7.64
6	2024	7	July	3003465.04	2559653.17	443811.87	14.78
7	2024	8	August	5497886.46	4809400.37	688486.09	12.52
8	2024	9	September	4056998.01	3429889.12	627108.89	15.46
9	2024	10	October	3447701.27	3232547.66	215153.61	6.24
10	2024	11	November	5702508.12	4953264.07	749244.05	13.14
11	2024	12	December	7830553.97	6657466.38	1173087.59	14.98
12	2025	1	January	8414196.75	7305592.27	1108604.48	13.18
13	2025	2	February	7245577.43	6262923.91	982653.52	13.56
14	2025	3	March	9850846.41	8722381.30	1128465.11	11.46
15	2025	4	April	12845414.00	11256824.30	1588589.70	12.37
16	2025	5	May	7787369.36	6684595.78	1102773.58	14.16

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Monthly sales and profit can vary over time. Compare the actual monthly results and profit margins to identify periods of stronger or weaker profitability.

SUGGESTED BUSINESS ACTION

Investigate the causes of the observed changes before recommending action.

OUTPUT FILE `SQL_Outputs/Q01_Monthly_Sales_Profit.csv`

Q1. What is the monthly sales and profit trend?

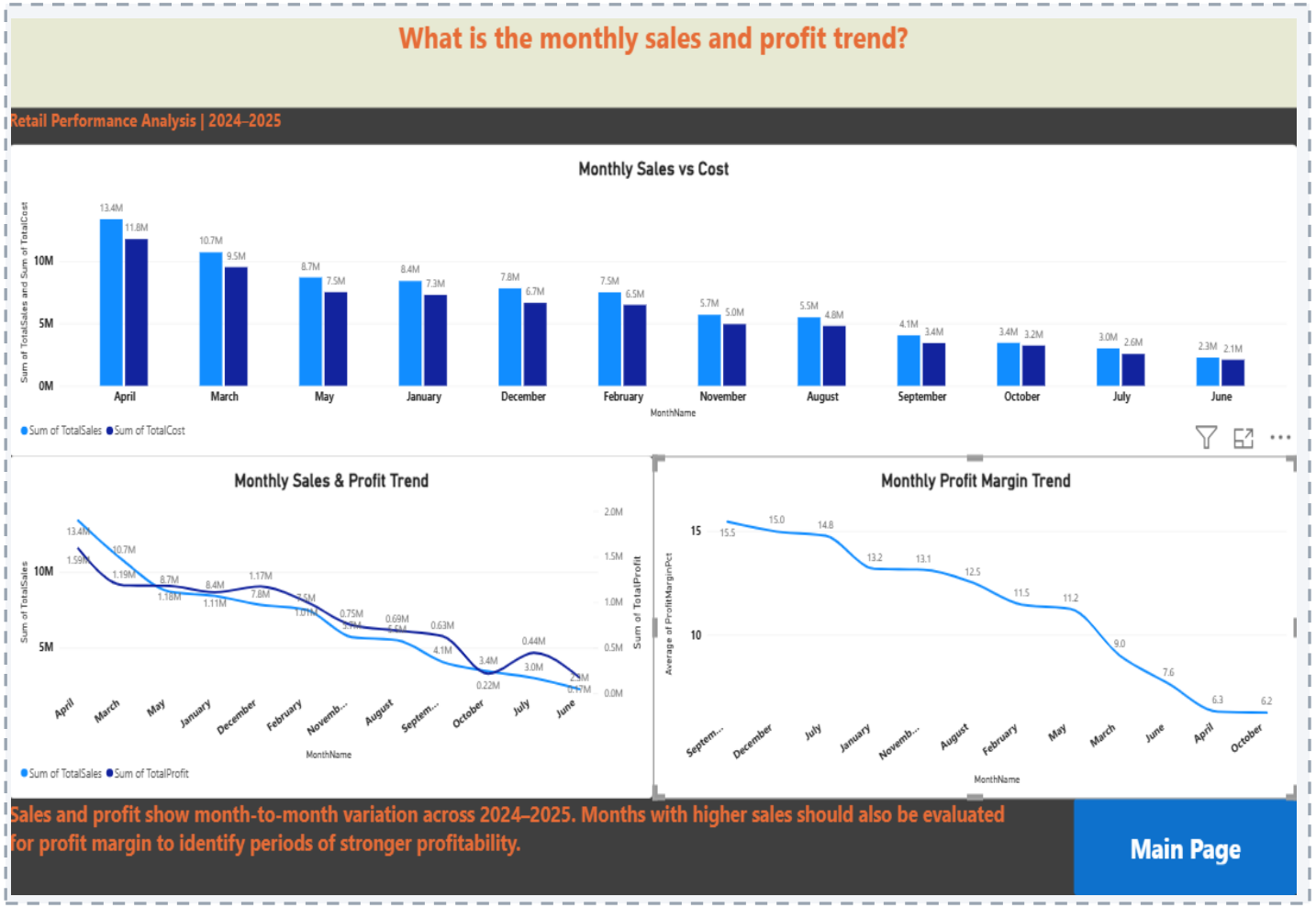


CHART PLACEHOLDERS

1. **Monthly Sales & Profit Trend**
2. **Monthly Sales vs Cost**
3. **Monthly Profit Margin Trend**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Monthly sales and profit can vary over time. Compare the actual monthly results and profit margins to identify periods of stronger or weaker profitability.

SUGGESTED BUSINESS ACTION

Investigate the causes of the observed changes before recommending action.

Q2. Which are the top 10 products by revenue?

BUSINESS QUESTION

Which are the top 10 products by revenue?

BUSINESS OBJECTIVE

Identify which products contribute the most revenue.

SQL QUERY

```
SELECT TOP 10
  p.ProductID,
  p.ProductName,
  p.Category,
  p.SubCategory,
  SUM(f.Quantity) AS UnitsSold,
  SUM(f.SalesAmount) AS Revenue,
  SUM(f.ProfitAmount) AS Profit,
  CAST(
    100.0 * SUM(f.ProfitAmount)
    / NULLIF(SUM(f.SalesAmount),0)
    AS DECIMAL(10,2)
  ) AS ProfitMarginPct
FROM dbo.FactSales f
JOIN dbo.DimProduct p
  ON f.ProductID = p.ProductID
GROUP BY
  p.ProductID,
  p.ProductName,
  p.Category,
  p.SubCategory
ORDER BY Revenue DESC;
```

	ProductID	ProductName	Category	SubCategory	UnitsSold	Revenue	Profit	ProfitMarginPct
1	437	TechPro Appliances Product 437	Home	Appliances	27	1603034.47	506972.17	31.63
2	428	StyleCo Accessories Product 428	Fashion	Accessories	23	1331088.93	221311.33	16.63
3	492	HomePlus Accessories Product 492	Electronics	Accessories	20	1164358.70	194723.30	16.72
4	233	FitMax Audio Product 233	Electronics	Audio	23	1127737.29	261402.50	23.18
5	256	StyleCo Training Product 256	Sports	Training	22	960931.79	-58215.13	-6.06
6	119	FitMax Footwear Product 119	Fashion	Footwear	16	947806.21	174907.49	18.45
7	161	FitMax Footwear Product 161	Fashion	Footwear	25	877550.31	292226.56	33.30
8	226	Prime Footwear Product 226	Fashion	Footwear	25	816120.55	101937.80	12.49
9	12	Apex Accessories Product 012	Electronics	Accessories	26	814583.36	-165729.48	-20.35
10	57	FreshMart Appliances Product 057	Home	Appliances	18	811452.24	55065.42	6.79

SQL CONCEPTS USED

Aggregation (SUM) – Calculates total revenue for each product.
GROUP BY – Groups records by product.
ORDER BY – Ranks products by total revenue in descending order.
TOP 10 – Limits the result to the 10 products with the highest revenue.
Use only the concepts above if they appear in your actual SQL query.

Q2. Which are the top 10 products by revenue?

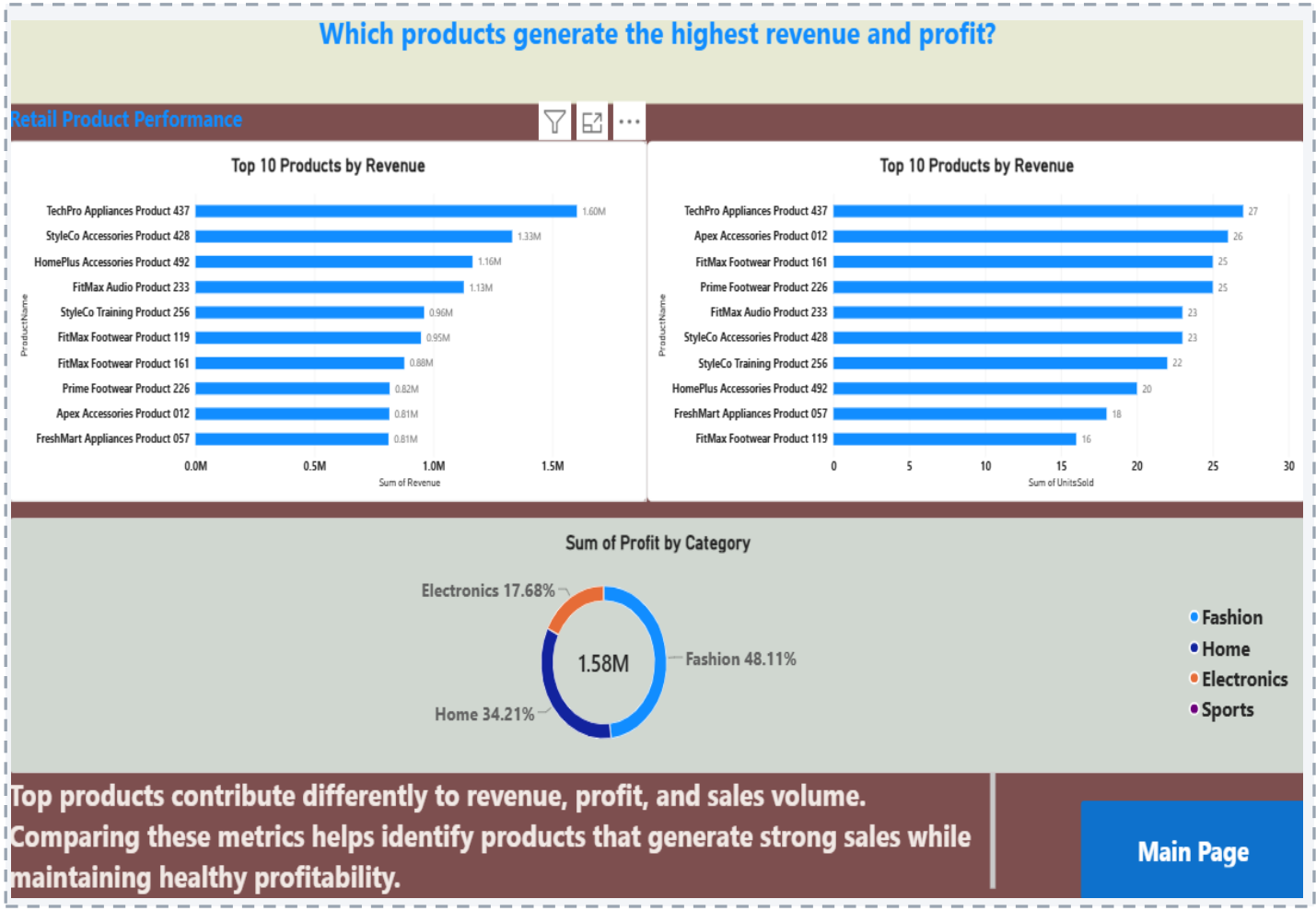


CHART PLACEHOLDERS

1. **Top 10 Products by Revenue**
2. **Top 10 Products by Profit**
3. **Top 10 Products by Units Sold**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Comparing revenue, profit, and units sold can reveal whether the best-selling products are also the most profitable. Confirm the result from the actual output.

SUGGESTED BUSINESS ACTION

Confirm the ranking in the actual output before drawing product conclusions.

Q3. How can customers be segmented based on purchasing behavior?

BUSINESS QUESTION

How can customers be segmented based on purchasing behavior?

BUSINESS OBJECTIVE

Group customers by purchasing behavior to support targeted retention and engagement.

SQL QUERY

```sql
WITH CustomerRFM AS
(
    SELECT
        c.CustomerID,
        c.CustomerName,
        c.Region,
        MAX(d.FullDate) AS LastPurchaseDate,
        DATEDIFF(
            DAY,
            MAX(d.FullDate),
            (SELECT MAX(FullDate) FROM dbo.DimDate)
        ) AS RecencyDays,
        COUNT(DISTINCT f.SaleID) AS Frequency,
        SUM(f.SalesAmount) AS MonetaryValue
    FROM dbo.DimCustomer c
    JOIN dbo.FactSales f
        ON c.CustomerID = f.CustomerID
    JOIN dbo.DimDate d
        ON f.DateID = d.DateID
    GROUP BY
        c.CustomerID,
        c.CustomerName,
        c.Region
),
Scored AS
(
    SELECT *,
        NTILE(4) OVER (ORDER BY RecencyDays DESC) AS RecencyScore,
        NTILE(4) OVER (ORDER BY Frequency) AS FrequencyScore,
        NTILE(4) OVER (ORDER BY MonetaryValue) AS MonetaryScore
    FROM CustomerRFM
)
SELECT
    CustomerID,
    CustomerName,
    Region,
    RecencyDays,
    Frequency,
    MonetaryValue,
    RecencyScore,
    FrequencyScore,
    MonetaryScore,
    CASE
        WHEN RecencyScore >= 3
         AND FrequencyScore >= 3
         AND MonetaryScore >= 3
            THEN 'VIP / High Value'
        WHEN MonetaryScore >= 3
         AND FrequencyScore >= 2
            THEN 'Loyal'
        WHEN RecencyScore <= 2
         AND FrequencyScore >= 2
            THEN 'At Risk'
        WHEN RecencyScore <= 2
         AND FrequencyScore <= 2
            THEN 'Lost / Inactive'
        ELSE 'Regular'
    END AS CustomerSegment
FROM Scored;
```

	CustomerID	CustomerName	Region	RecencyDays	Frequency	MonetaryValue	RecencyScore	FrequencyScore	MonetaryScore	CustomerSegment
1	27	Vivaan Jadhav ...	West	73	1	512.78	3	2	1	Regular
2	202	Rohan Sharma 2...	West	61	1	658.28	3	2	1	Regular
3	3	Priya Joshi 003	South	43	1	1041.06	3	2	1	Regular
4	40	Neel Patil 040	West	168	1	2300.04	2	1	1	Lost / Inactive
5	296	Aditya Singh 296	West	229	1	2550.04	1	1	1	Lost / Inactive
6	401	Aditi Ade 401	South	56	1	2708.21	3	2	1	Regular
7	31	Yash Jadhav 031	Cent...	36	1	5458.47	4	2	1	Regular
8	56	Kavya Kadam 056	Cent...	322	1	6238.86	1	1	1	Lost / Inactive
9	482	Arjun Jadhav 482	West	304	1	6867.91	1	1	1	Lost / Inactive
10	457	Pooja Kulkarni 457	West	9	1	8861.02	4	3	1	Regular
11	269	Meera More 269	West	98	1	11935.47	2	2	1	At Risk
12	238	Rohan Patil 238	West	106	1	12317.65	2	2	1	At Risk
13	231	Arjun Kulkarni 231	Cent...	308	1	12577.48	1	1	1	Lost / Inactive
14	39	Yash Ade 039	South	10	1	13810.28	4	3	1	Regular
15	185	Yash Deshmukh...	West	56	1	13916.06	3	2	1	Regular
16	261	Rahul Ade 261	West	4	1	14358.84	4	3	1	Regular
17	100	Meera Mishra 100	South	48	1	16095.64	3	2	1	Regular
18	414	Vikram Verma 414	West	12	1	16817.40	4	3	1	Regular
19	156	Yash Ade 156	West	167	1	17982.83	2	1	1	Lost / Inactive
20	278	Aditi Verma 278	Cent...	429	1	20010.54	1	1	1	Lost / Inactive
21	489	Rahul Patil 489	Cent...	10	1	20621.34	4	3	1	Regular
22	463	Vivaan Sharma ...	West	276	1	22515.65	1	1	1	Lost / Inactive
23	7	Aditya Kadam 007	West	141	1	23205.81	2	1	1	Lost / Inactive
24	371	Priya Jadhav 371	South	65	1	25315.38	3	2	1	Regular
25	361	Arjun Kadam 361	West	47	1	26148.84	3	2	1	Regular
26	446	Sneha Deshmuk...	South	109	1	26285.41	2	2	1	At Risk
27	441	Aditi Verma 441	Cent...	39	1	26969.46	3	2	1	Regular
28	498	Ananya Patil 498	West	16	1	27137.21	4	3	1	Regular
29	483	Rahul Patil 483	Cent...	37	1	28112.15	4	2	1	Regular
30	419	Sneha Singh 419	Cent...	307	1	29030.99	1	1	1	Lost / Inactive
31	418	Vikram Singh 418	West	292	1	31375.70	1	1	1	Lost / Inactive
32	430	Aditya Verma 430	West	100	1	31384.78	2	2	1	At Risk
33	83	Riya Joshi 083	West	77	1	32305.67	3	2	1	Regular
34	46	Aditi Singh 046	Cent...	28	1	32479.28	4	2	1	Regular
35	328	Pooja Deshmuk...	South	30	1	34904.15	4	2	1	Regular
36	97	Aditi More 097	Cent...	148	2	36148.05	2	4	1	At Risk
37	63	Aditi Kadam 063	Cent...	67	1	38127.26	3	2	1	Regular
38	262	Rahul Deshmuk...	West	82	1	38732.86	3	2	1	Regular
39	242	Ananya Verma 2...	West	468	1	40328.40	1	1	1	Lost / Inactive
40	59	Rohan Sharma 0...	West	39	1	40520.26	3	2	1	Regular

SQL CONCEPTS USED

Explain customer segmentation or RFM only to the extent implemented in your query.

List only concepts that appear in your actual query.

OUTPUT FILE

`SQL_Outputs/Q03_Customer_Segmentation.csv`

Q3. How can customers be segmented based on purchasing behavior?

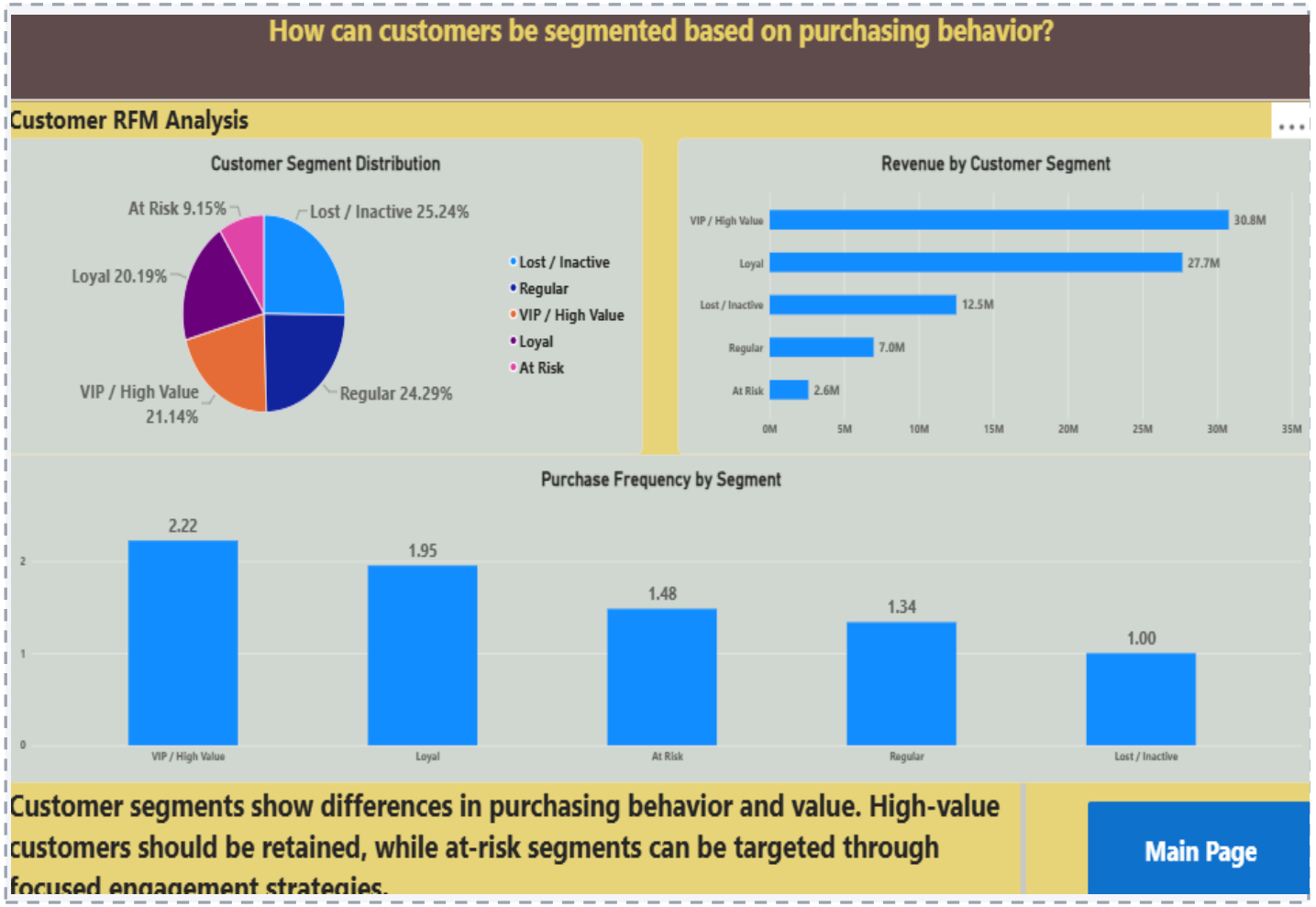


CHART PLACEHOLDERS

1. Customer Segment Distribution
2. Revenue by Customer Segment
3. Purchase Frequency by Segment

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Customer segments may differ in purchasing frequency and monetary value. Use the actual results to determine which groups need retention or re-engagement strategies.

SUGGESTED BUSINESS ACTION

Define retention or re-engagement actions only after reviewing the actual segments.

Q4. Which regions perform best in sales and profitability?

BUSINESS QUESTION

Which regions perform best in sales and profitability?

BUSINESS OBJECTIVE

Compare regional sales and profitability.

SQL QUERY

```
SELECT
  c.Region,
  COUNT(DISTINCT f.CustomerID) AS ActiveCustomers,
  COUNT(DISTINCT f.SaleID) AS Orders,
  SUM(f.Quantity) AS UnitsSold,
  SUM(f.SalesAmount) AS Revenue,
  SUM(f.ProfitAmount) AS Profit,
  CAST(
    100.0 * SUM(f.ProfitAmount)
    / NULLIF(SUM(f.SalesAmount),0)
    AS DECIMAL(10,2)
  ) AS ProfitMarginPct
FROM dbo.FactSales f
JOIN dbo.DimCustomer c
  ON f.CustomerID = c.CustomerID
GROUP BY c.Region;
```

	Region	ActiveCustomers	Orders	UnitsSold	Revenue	Profit	ProfitMarginPct
1	Central	63	95	574	17421905.75	1986848.37	11.40
2	South	57	102	567	16521618.99	2152107.05	13.03
3	West	197	303	1638	46584760.00	6001391.18	12.88

SQL CONCEPTS USED

Explain how your query summarizes sales and profitability by region.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q04_Regional_Performance.csv`

Q4. Which regions perform best in sales and profitability?

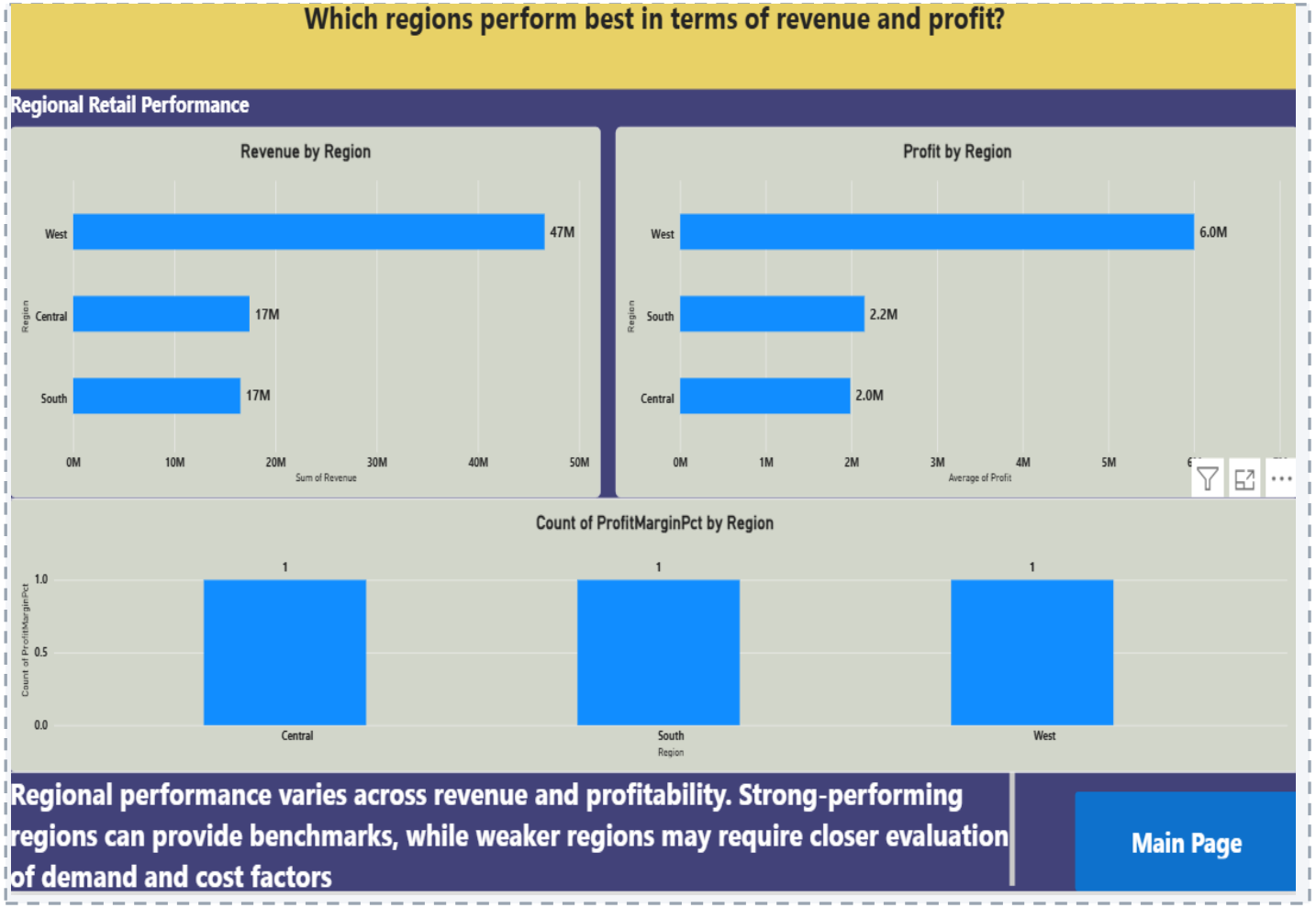


CHART PLACEHOLDERS

1. **Revenue by Region**
2. **Profit by Region**
3. **Profit Margin by Region**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Regional results can differ in revenue and profitability. Identify the actual highest- and lowest-performing regions before recommending business changes.

SUGGESTED BUSINESS ACTION

Identify the actual highest- and lowest-performing regions first.

Q5. What is the loan approval rate across different loan types?

BUSINESS QUESTION

What is the loan approval rate across different loan types?

BUSINESS OBJECTIVE

Measure loan approval performance across loan types.

SQL QUERY

```
SELECT
  LoanType,
  COUNT(*) AS TotalApplications,
  SUM(CASE WHEN LoanStatus = 'Approved' THEN 1 ELSE 0 END)
    AS ApprovedLoans,
  SUM(CASE WHEN LoanStatus = 'Rejected' THEN 1 ELSE 0 END)
    AS RejectedLoans,
  CAST(
    100.0 *
    SUM(CASE WHEN LoanStatus = 'Approved' THEN 1 ELSE 0 END)
    / NULLIF(COUNT(*),0)
    AS DECIMAL(10,2)
  ) AS ApprovalRatePct,
  SUM(
    CASE WHEN LoanStatus = 'Approved'
        THEN LoanAmount ELSE 0 END
  ) AS ApprovedLoanValue,
  AVG(
    CASE WHEN LoanStatus = 'Approved'
        THEN LoanAmount END
  ) AS AvgApprovedLoanAmount
FROM dbo.FactLoans
GROUP BY LoanType;
```

	LoanType	TotalApplications	ApprovedLoans	RejectedLoans	ApprovalRatePct	ApprovedLoanValue	AvgApprovedLoanAmount
1	Auto	82	52	16	63.41	38896481.40	748009.257692
2	Business	118	75	24	63.56	55473218.07	739642.907600
3	Education	84	56	17	66.67	44876731.69	801370.208750
4	Home	120	80	22	66.67	65927589.25	824094.865625
5	Personal	96	63	19	65.63	47932007.71	760825.519206

SQL CONCEPTS USED

Explain how your query calculates the approval rate for each loan type.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q05_Loan_Approval.csv`

Q5. What is the loan approval rate across different loan types?

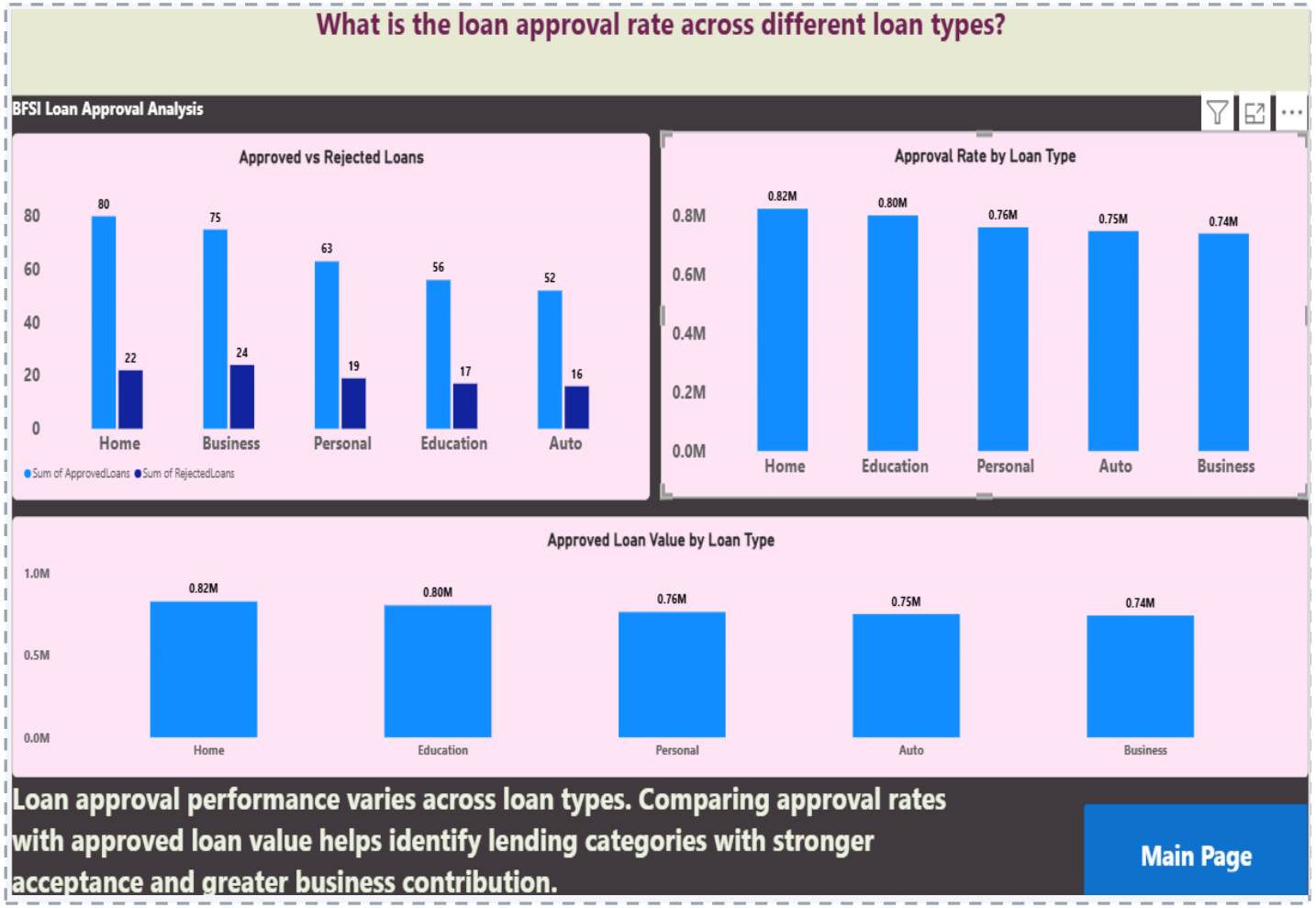


CHART PLACEHOLDERS

1. **Approved vs Rejected Loans**
2. **Approval Rate by Loan Type**
3. **Approved Loan Value by Loan Type**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare approval performance across loan types and evaluate approved loan value using actual results.

SUGGESTED BUSINESS ACTION

Validate how approval rate is defined before comparing loan types.

Q6. Which loan types and interest-rate bands have higher default risk?

BUSINESS QUESTION

Which loan types and interest-rate bands have higher default risk?

BUSINESS OBJECTIVE

Compare default risk across loan types and interest-rate bands.

SQL QUERY

```
SELECT
  l.LoanType,
  CASE
    WHEN l.InterestRate < 10 THEN 'Low Rate'
    WHEN l.InterestRate < 16 THEN 'Medium Rate'
    ELSE 'High Rate'
  END AS InterestRateBand,
  COUNT(*) AS TotalLoans,
  SUM(CAST(l.DefaultFlag AS INT)) AS Defaults,
  CAST(
    100.0 * SUM(CAST(l.DefaultFlag AS INT))
    / NULLIF(COUNT(*),0)
    AS DECIMAL(10,2)
  ) AS DefaultRatePct,
  SUM(l.LoanAmount) AS TotalLoanAmount,
  AVG(l.LoanAmount) AS AvgLoanAmount
FROM dbo.FactLoans l
WHERE l.LoanStatus = 'Approved'
GROUP BY
  l.LoanType,
  CASE
    WHEN l.InterestRate < 10 THEN 'Low Rate'
    WHEN l.InterestRate < 16 THEN 'Medium Rate'
    ELSE 'High Rate'
  END;
```

	LoanType	InterestRateBand	TotalLoans	Defaults	DefaultRatePct	TotalLoanAmount	AvgLoanAmount
1	Auto	High Rate	21	1	4.76	14979039.65	713287.602380
2	Business	High Rate	30	2	6.67	22906368.80	763545.626666
3	Education	High Rate	28	4	14.29	23799714.51	849989.803928
4	Home	High Rate	32	6	18.75	24788862.52	774651.953750
5	Personal	High Rate	21	0	0.00	18147346.13	864159.339523
6	Auto	Low Rate	13	1	7.69	9995699.14	768899.933846
7	Business	Low Rate	17	2	11.76	14211312.63	835959.566470
8	Education	Low Rate	5	0	0.00	4368524.55	873704.910000
9	Home	Low Rate	15	2	13.33	12788525.55	852568.370000
10	Personal	Low Rate	15	2	13.33	12336795.34	822453.022666
11	Auto	Medium Rate	18	3	16.67	13921742.61	773430.145000
12	Business	Medium Rate	28	4	14.29	18355536.64	655554.880000
13	Education	Medium Rate	23	2	8.70	16708492.63	726456.201304
14	Home	Medium Rate	33	3	9.09	28350201.18	859097.005454
15	Personal	Medium Rate	27	1	3.70	17447866.24	646217.268148

SQL CONCEPTS USED

Explain how your query defines default and groups loans into interest-rate bands.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q06_Loan_Default.csv`

Q6. Which loan types and interest-rate bands have higher default risk?

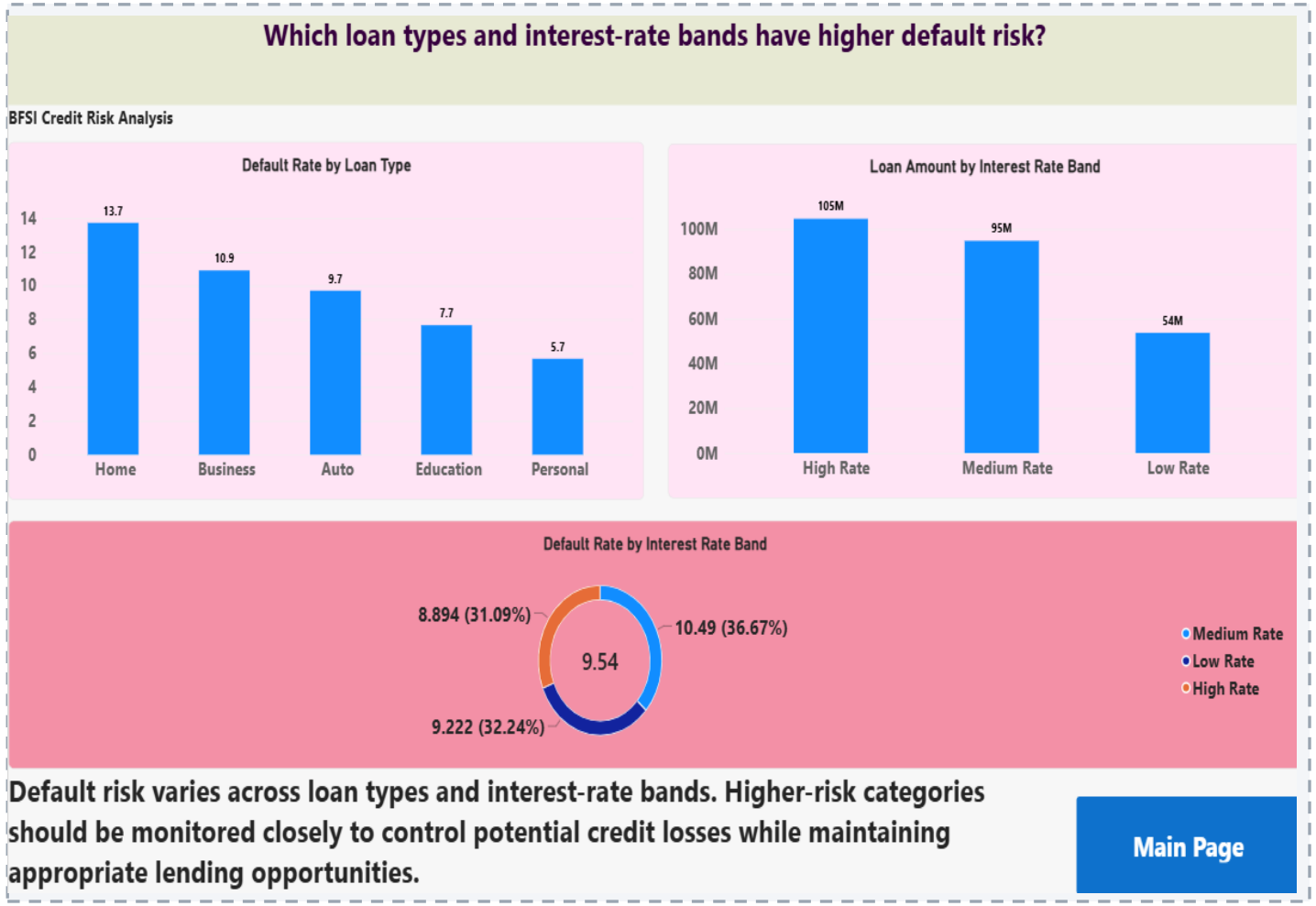


CHART PLACEHOLDERS

1. **Default Rate by Loan Type**
2. **Default Rate by Interest Rate Band**
3. **Loan Amount by Interest Rate Band**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare default rates across categories to identify where further risk monitoring may be useful. Do not assume that higher interest rates cause defaults.

SUGGESTED BUSINESS ACTION

Treat differences as associations, not causes; validate before any risk decision.

Q7. What are the major transaction trends by type and status?

BUSINESS QUESTION

What are the major transaction trends by type and status?

BUSINESS OBJECTIVE

Understand transaction volumes and values by type and status.

SQL QUERY —

```
SELECT
  l.LoanType,
  CASE
    WHEN l.InterestRate < 10 THEN 'Low Rate'
    WHEN l.InterestRate < 16 THEN 'Medium Rate'
    ELSE 'High Rate'
  END AS InterestRateBand,
  COUNT(*) AS TotalLoans,
  SUM(CAST(l.DefaultFlag AS INT)) AS Defaults,
  CAST(
    100.0 * SUM(CAST(l.DefaultFlag AS INT))
    / NULLIF(COUNT(*),0)
    AS DECIMAL(10,2)
  ) AS DefaultRatePct,
  SUM(l.LoanAmount) AS TotalLoanAmount,
  AVG(l.LoanAmount) AS AvgLoanAmount
FROM dbo.FactLoans l
WHERE l.LoanStatus = 'Approved'
GROUP BY
  l.LoanType,
  CASE
    WHEN l.InterestRate < 10 THEN 'Low Rate'
    WHEN l.InterestRate < 16 THEN 'Medium Rate'
    ELSE 'High Rate'
  END;
```

	LoanType	InterestRateBand	TotalLoans	Defaults	DefaultRatePct	TotalLoanAmount	AvgLoanAmount
1	Auto	High Rate	21	1	4.76	14979039.65	713287.602380
2	Business	High Rate	30	2	6.67	22906368.80	763545.626666
3	Education	High Rate	28	4	14.29	23799714.51	849989.803928
4	Home	High Rate	32	6	18.75	24788862.52	774651.953750
5	Personal	High Rate	21	0	0.00	18147346.13	864159.339523
6	Auto	Low Rate	13	1	7.69	9995699.14	768899.933846
7	Business	Low Rate	17	2	11.76	14211312.63	835959.566470
8	Education	Low Rate	5	0	0.00	4368524.55	873704.910000
9	Home	Low Rate	15	2	13.33	12788525.55	852568.370000
10	Personal	Low Rate	15	2	13.33	12336795.34	822453.022666
11	Auto	Medium Rate	18	3	16.67	13921742.61	773430.145000
12	Business	Medium Rate	28	4	14.29	18355536.64	655554.880000
13	Education	Medium Rate	23	2	8.70	16708492.63	726456.201304
14	Home	Medium Rate	33	3	9.09	28350201.18	859097.005454
15	Personal	Medium Rate	27	1	3.70	17447866.24	646217.268148

SQL CONCEPTS USED

Explain how your query summarizes transactions by type, status and time.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q07_Transaction_Analysis.csv`

Q7. What are the major transaction trends by type and status?

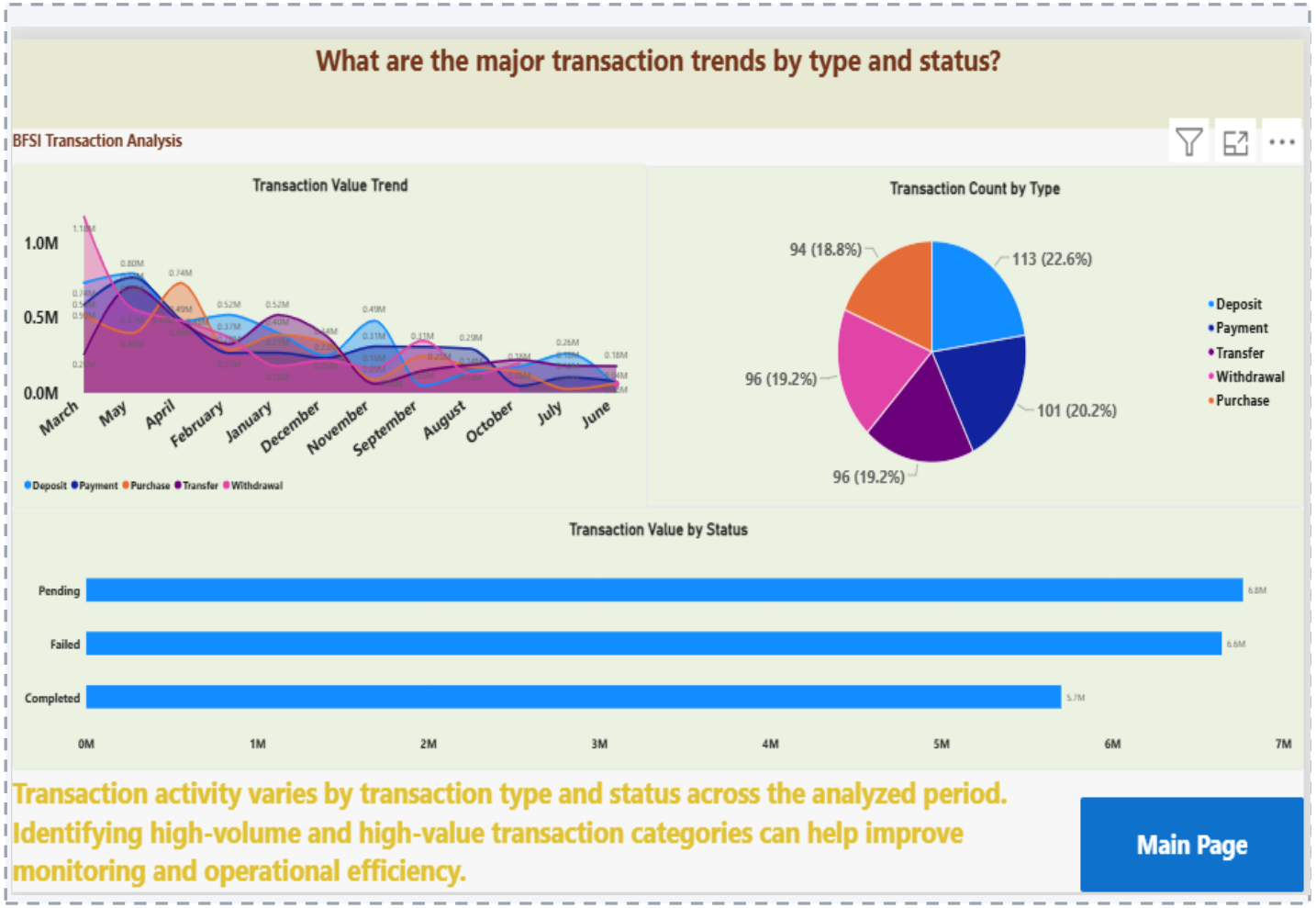


CHART PLACEHOLDERS

1. **Transaction Value Trend**
2. **Transaction Count by Type**
3. **Transaction Value by Status**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare transaction volumes and values by type and status to identify operational patterns and categories that may require investigation.

SUGGESTED BUSINESS ACTION

Investigate unusual categories in the actual data before acting.

Q8. Which customers have the highest financial value?

BUSINESS QUESTION

Which customers have the highest financial value?

BUSINESS OBJECTIVE

Identify high-value customers and separately assess their default exposure.

SQL QUERY

```sql
WITH TransactionValue AS
(
    SELECT
        CustomerID,
        SUM(Amount) AS TotalTransactionValue,
        COUNT(*) AS TransactionCount
    FROM dbo.FactTransactions
    WHERE TransactionStatus = 'Completed'
    GROUP BY CustomerID
),
LoanValue AS
(
    SELECT
        CustomerID,
        SUM(
            CASE
                WHEN LoanStatus = 'Approved'
                THEN LoanAmount
                ELSE 0
            END
        ) AS ApprovedLoanValue,
        SUM(
            CASE
                WHEN LoanStatus = 'Approved'
                THEN 1
                ELSE 0
            END
        ) AS ApprovedLoanCount,
        SUM(CAST(DefaultFlag AS INT)) AS DefaultCount
    FROM dbo.FactLoans
    GROUP BY CustomerID
)
SELECT
    c.CustomerID,
    c.CustomerName,
    c.Region,
    c.IncomeBand,
    ISNULL(t.TotalTransactionValue,0) AS TotalTransactionValue,
    ISNULL(t.TransactionCount,0) AS TransactionCount,
    ISNULL(l.ApprovedLoanValue,0) AS ApprovedLoanValue,
    ISNULL(l.ApprovedLoanCount,0) AS ApprovedLoanCount,
    ISNULL(l.DefaultCount,0) AS DefaultCount,
    ISNULL(t.TotalTransactionValue,0)
        + ISNULL(l.ApprovedLoanValue,0) AS FinancialValue
FROM dbo.DimCustomer c
LEFT JOIN TransactionValue t
    ON c.CustomerID = t.CustomerID
LEFT JOIN LoanValue l
    ON c.CustomerID = l.CustomerID
```

	CustomerID	CustomerName	Region	IncomeBand	TotalTransactionValue	TransactionCount	ApprovedLoanValue	ApprovedLoanCount	DefaultCount	FinancialValue
1	1	Vikram Deshmukh 001	West	High	0.00	0	1831080.02	2	0	1831080.02
2	3	Priya Joshi 003	South	Low	0.00	0	1594077.14	2	1	1594077.14
3	4	Rohan Joshi 004	West	Low	65424.02	2	858816.98	1	0	924241.00
4	6	Aditi Mishra 006	West	High	31742.03	1	0.00	0	0	31742.03
5	7	Aditya Kadam 007	West	Medium	0.00	0	699253.75	1	0	699253.75
6	8	Rahul Pawar 008	West	High	16939.41	1	0.00	0	0	16939.41
7	9	Aditi Verma 009	West	Low	0.00	0	1119899.42	1	0	1119899.42
8	10	Vikram Verma 010	South	Low	24246.58	1	0.00	0	0	24246.58
9	11	Karan Mishra 011	West	Low	45404.90	1	1003312.89	1	0	1048717.79
10	12	Neel Deshmukh 012	West	High	18297.78	1	773367.84	1	0	791665.62
11	13	Rohan Chavan 013	West	Low	0.00	0	0.00	0	0	0.00
12	14	Aditi Patil 014	Central	High	62510.27	1	0.00	0	0	62510.27
13	15	Aarav Verma 015	South	High	0.00	0	0.00	0	0	0.00
14	16	Arjun Kulkarni 016	West	Medium	0.00	0	303783.19	1	0	303783.19
15	17	Rahul Chavan 017	South	High	49409.08	1	1015784.44	1	0	1065193.52
16	21	Aditi Jadhav 021	South	High	28022.37	1	2220022.00	3	1	2248044.37
17	22	Sneha Mishra 022	West	Low	0.00	0	0.00	0	0	0.00
18	23	Meera Chavan 023	West	Low	0.00	0	0.00	0	0	0.00
19	24	Aditya Mishra 024	West	Low	0.00	0	1251775.43	1	0	1251775.43
20	25	Karan Chavan 025	Central	Low	0.00	0	278236.60	1	0	278236.60
21	26	Priya Pawar 026	West	Medium	0.00	0	2394464.24	3	0	2394464.24
22	27	Vivaan Jadhav 027	West	Low	25392.47	1	0.00	0	0	25392.47
23	29	Arjun Sharma 029	South	Low	68858.36	1	156086.06	1	0	224944.42
24	30	Karan Kadam 030	Central	Low	0.00	0	1312720.31	1	0	1312720.31
25	31	Yash Jadhav 031	Central	Low	0.00	0	232859.13	1	0	232859.13
26	32	Meera Singh 032	West	High	0.00	0	844741.40	1	0	844741.40
27	34	Aditi Patil 034	West	Medium	110635.05	2	0.00	0	0	110635.05
28	35	Vivaan Ade 035	South	High	0.00	0	484854.54	1	0	484854.54
29	39	Yash Ade 039	South	High	0.00	0	119478.42	1	0	119478.42
30	40	Neel Patil 040	West	Medium	7812.92	1	0.00	0	0	7812.92
31	44	Kavya Sharma 044	West	Medium	0.00	0	762151.81	1	0	762151.81

SQL CONCEPTS USED

Explain how your query defines and combines customer financial value.

List only concepts that appear in your actual query.

OUTPUT FILE

`SQL_Outputs/Q08_Customer_Financial_Value.csv`

Q8. Which customers have the highest financial value?

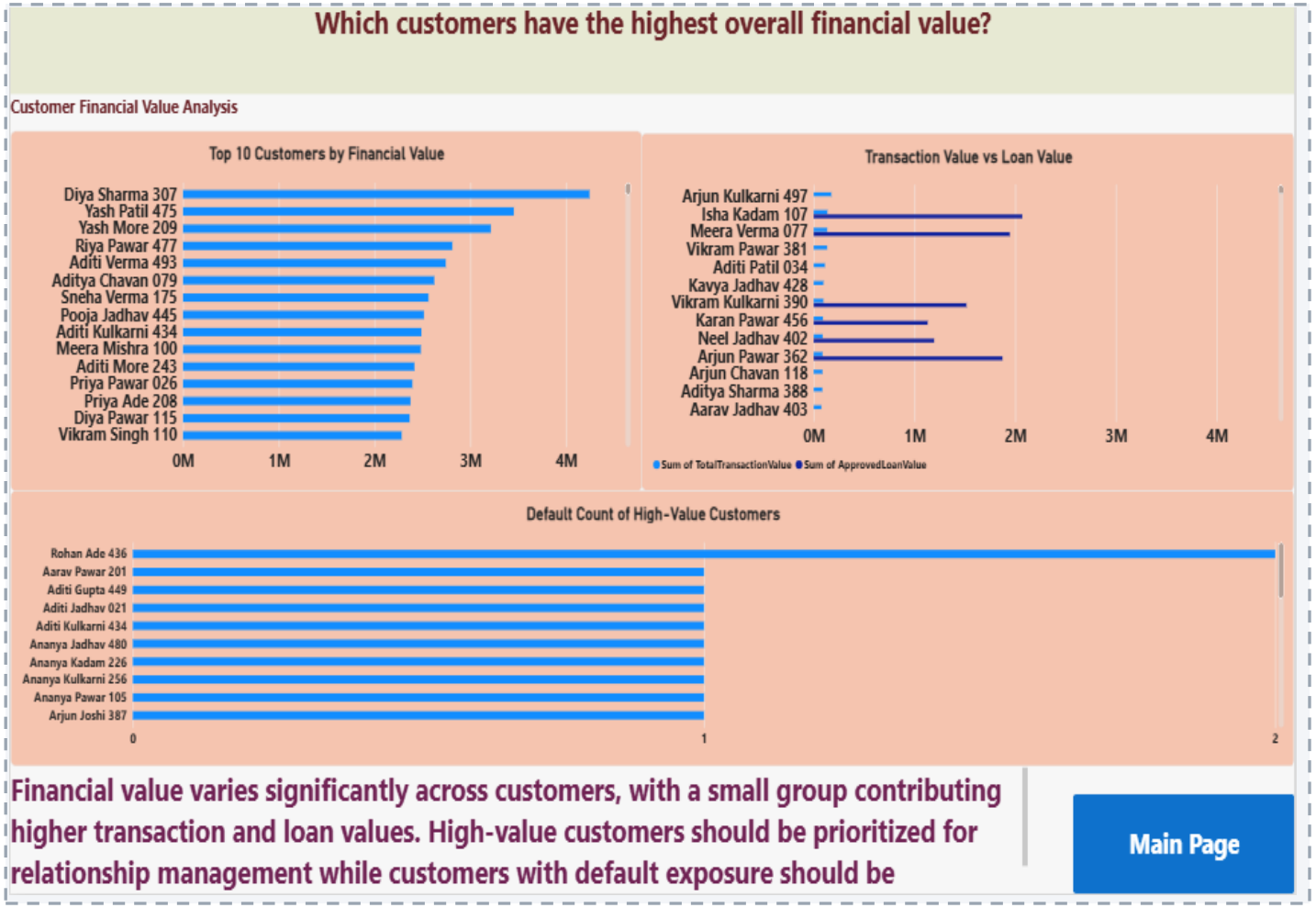


CHART PLACEHOLDERS

1. **Top 10 Customers by Financial Value**
2. **Transaction Value vs Approved Loan Value**
3. **Default Count by Customer**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Customer financial value may vary according to transaction activity and approved lending. Use the actual output to identify high-value customers and separately assess default exposure.

SUGGESTED BUSINESS ACTION

Assess default exposure separately from customer value.

Q9. How do patient visits change over time?

BUSINESS QUESTION

How do patient visits change over time?

BUSINESS OBJECTIVE

Track how visits, visit status and treatment cost change over time.

SQL QUERY

```
SELECT
  d.Year,
  d.Month,
  d.MonthName,
  COUNT(*) AS TotalVisits,
  SUM(
    CASE WHEN v.VisitStatus = 'Completed'
      THEN 1 ELSE 0 END
  ) AS CompletedVisits,
  SUM(
    CASE WHEN v.VisitStatus = 'Cancelled'
      THEN 1 ELSE 0 END
  ) AS CancelledVisits,
  SUM(
    CASE WHEN v.VisitStatus = 'No Show'
      THEN 1 ELSE 0 END
  ) AS NoShowVisits,
  SUM(v.TreatmentCost) AS TotalTreatmentCost
FROM dbo.FactHealthcareVisits v
JOIN dbo.DimDate d
  ON v.DateID = d.DateID
GROUP BY
  d.Year,
  d.Month,
  d.MonthName;
```

	Year	Month	MonthName	TotalVisits	CompletedVisits	CancelledVisits	NoShowVisits	TotalTreatmentCos
1	2024	2	February	4	3	0	1	19615.83
2	2024	3	March	5	3	1	1	20092.89
3	2024	4	April	6	4	1	1	21437.83
4	2024	5	May	8	6	1	1	40921.81
5	2024	6	June	10	8	1	1	39917.97
6	2024	7	July	15	9	1	5	54870.19
7	2024	8	August	18	16	2	0	85739.00
8	2024	9	September	22	17	3	2	105272.56
9	2024	10	October	28	24	1	3	146559.68
10	2024	11	November	39	33	3	3	193879.66
11	2024	12	December	33	26	5	2	185464.09
12	2025	1	January	48	39	5	4	225407.70
13	2025	2	February	52	38	6	8	245075.93
14	2025	3	March	74	63	6	5	424256.31
15	2025	4	April	74	60	8	6	392191.32
16	2025	5	May	64	51	7	6	315033.11

SQL CONCEPTS USED

Explain how your query summarizes visits, statuses and costs over time.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q09_Patient_Visits.csv`

Q9. How do patient visits change over time?

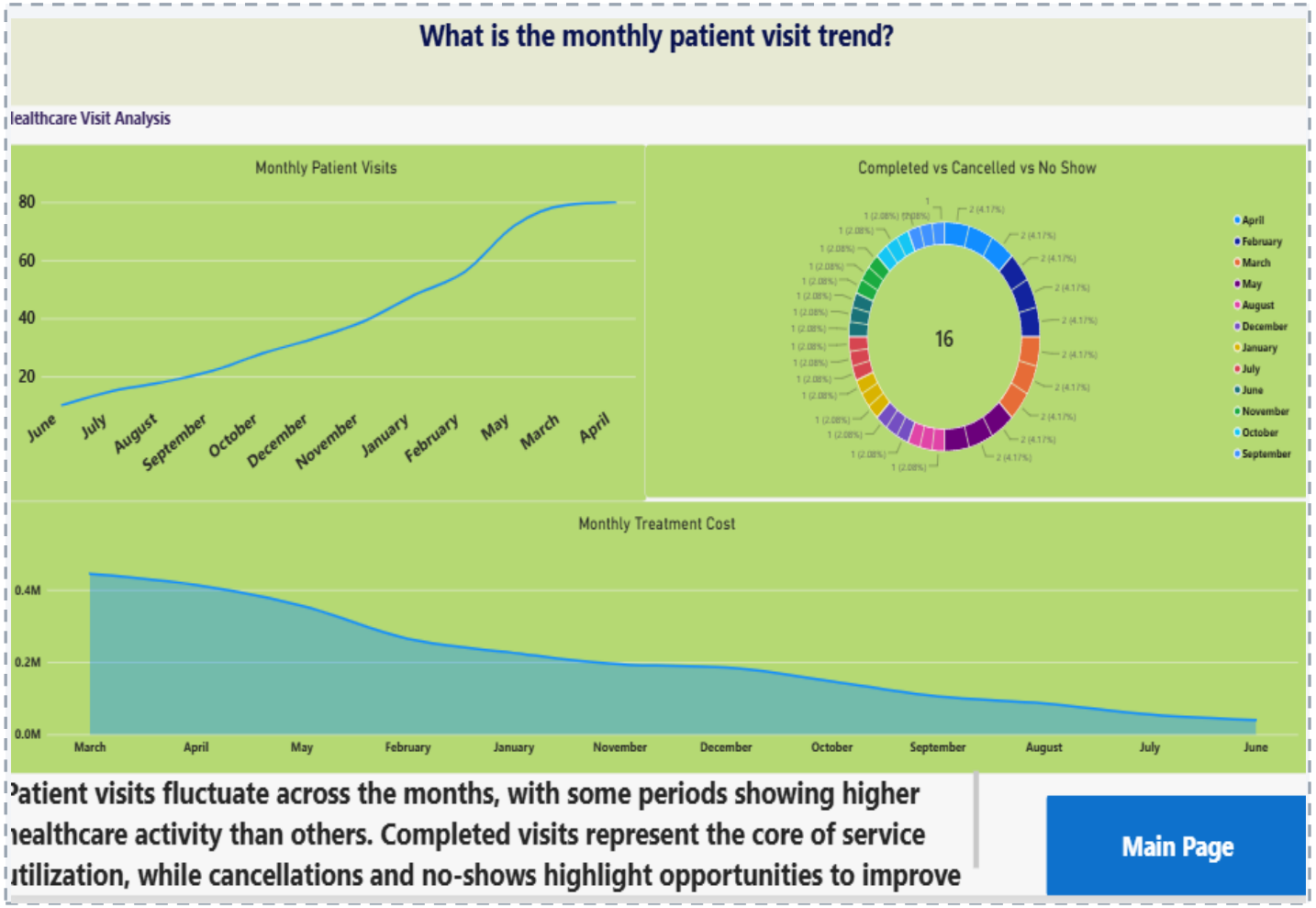


CHART PLACEHOLDERS

1. **Monthly Patient Visits**
2. **Completed vs Cancelled vs No-Show Visits**
3. **Monthly Treatment Cost**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare visit volume, visit status, and treatment costs over time to identify actual changes in service utilization.

SUGGESTED BUSINESS ACTION

Validate the observed trends before planning any capacity changes.

Q10. Which healthcare departments have the strongest performance?

BUSINESS QUESTION

Which healthcare departments have the strongest performance?

BUSINESS OBJECTIVE

Compare departments on visit volume, treatment cost and completion rate.

SQL QUERY — PLACEHOLDER

```
SELECT

dp.DepartmentID,

  dp.DepartmentName,
  dp.DepartmentType,
  COUNT(*) AS TotalVisits,
  SUM(
    CASE WHEN v.VisitStatus = 'Completed'
        THEN 1 ELSE 0 END
  ) AS CompletedVisits,
  SUM(v.TreatmentCost) AS TotalTreatmentCost,
  AVG(v.TreatmentCost) AS AvgTreatmentCost,
  CAST(
    100.0 *
    SUM(
      CASE WHEN v.VisitStatus = 'Completed'
          THEN 1 ELSE 0 END
    ) / NULLIF(COUNT(*),0)
    AS DECIMAL(10,2) ) AS CompletionRatePct
FROM dbo.FactHealthcareVisits v
JOIN dbo.DimDepartment dpON v.DepartmentID = dp.DepartmentID
GROUP BY
  dp.DepartmentID,
  dp.DepartmentName,
  dp.DepartmentType;
```

	DepartmentID	DepartmentName	DepartmentType	TotalVisits	CompletedVisits	TotalTreatmentCost	AvgTreatmentCost	CompletionRatePct
1	1	Clinical Department 001	Clinical	1	1	1656.57	1656.570000	100.00
2	2	Diagnostic Departmen...	Diagnostic	1	1	11937.73	11937.730000	100.00
3	3	Surgical Department 0...	Surgical	2	1	5487.95	2743.975000	50.00
4	4	Support Department 0...	Support	1	1	2113.23	2113.230000	100.00
5	6	Diagnostic Departmen...	Diagnostic	1	1	7016.88	7016.880000	100.00
6	7	Surgical Department 0...	Surgical	1	1	9713.76	9713.760000	100.00
7	8	Support Department 0...	Support	3	1	5167.42	1722.473333	33.33
8	10	Diagnostic Departmen...	Diagnostic	1	1	11343.40	11343.400000	100.00
9	12	Support Department 0...	Support	1	1	591.07	591.070000	100.00
10	13	Clinical Department 013	Clinical	1	1	8347.81	8347.810000	100.00
11	15	Surgical Department 0...	Surgical	1	0	0.00	0.000000	0.00
12	16	Support Department 0...	Support	1	0	0.00	0.000000	0.00
13	17	Clinical Department 017	Clinical	4	4	23425.55	5856.387500	100.00
14	18	Diagnostic Departmen...	Diagnostic	1	0	0.00	0.000000	0.00
15	20	Support Department 0...	Support	1	1	1933.83	1933.830000	100.00
16	22	Diagnostic Departmen...	Diagnostic	2	2	8190.69	4095.345000	100.00
17	24	Support Department 0...	Support	1	0	0.00	0.000000	0.00
18	25	Clinical Department 025	Clinical	1	1	6929.47	6929.470000	100.00
19	26	Diagnostic Departmen...	Diagnostic	1	0	0.00	0.000000	0.00
20	27	Surgical Department 0...	Surgical	1	1	5552.49	5552.490000	100.00
21	28	Support Department 0...	Support	1	1	6410.01	6410.010000	100.00
22	30	Diagnostic Departmen...	Diagnostic	1	1	7997.42	7997.420000	100.00

SQL CONCEPTS USED

Explain how your query defines completion rate and summarizes each department.

List only concepts that appear in your actual query.

OUTPUT FILE

`SQL_Outputs/Q10_Department_Performance.csv`

Q10. Which healthcare departments have the strongest performance?

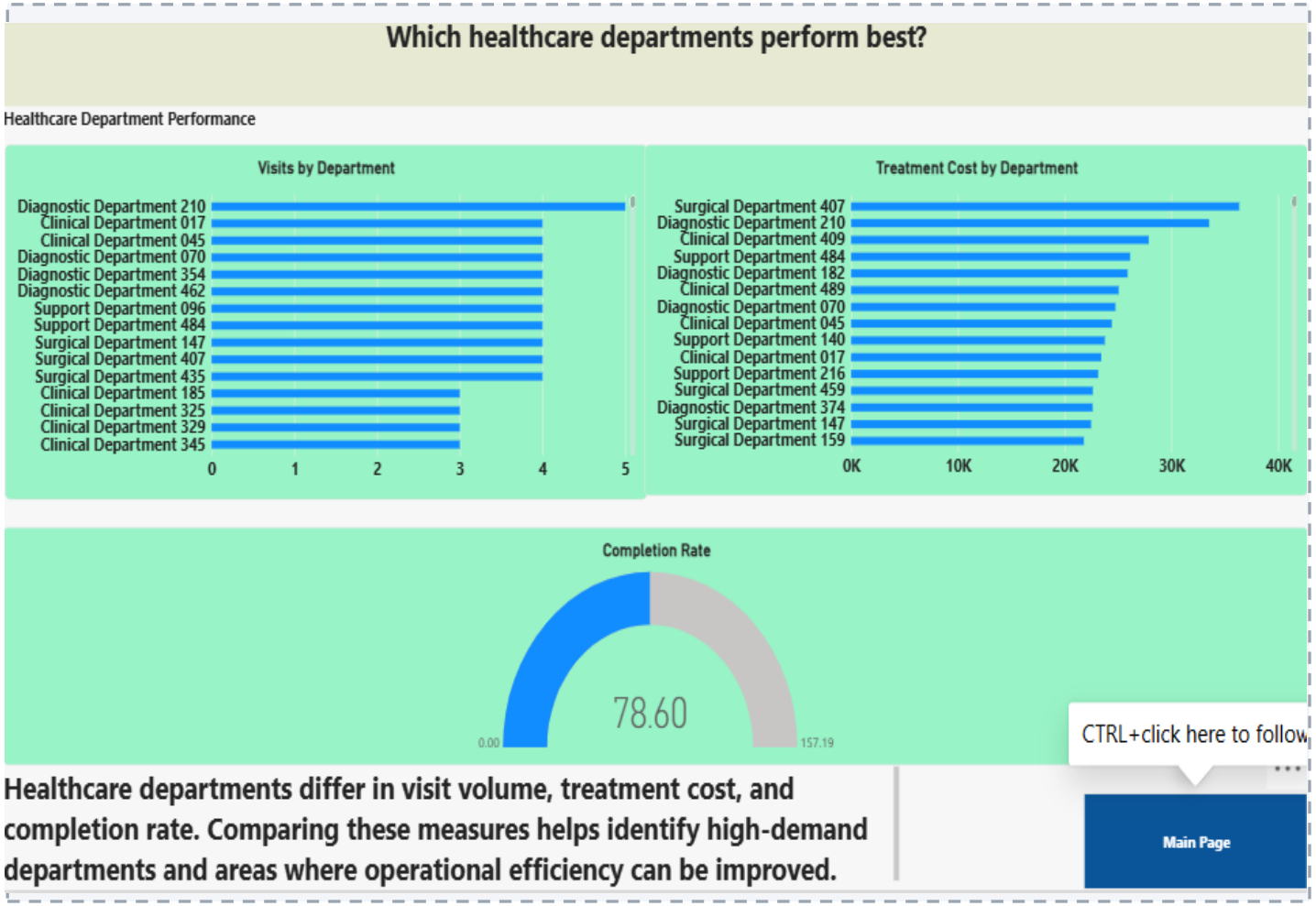


CHART PLACEHOLDERS

1. **Visits by Department**
2. **Treatment Cost by Department**
3. **Completion Rate by Department**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare departments on visit volume, treatment cost, and completion rate. Avoid calling a department more efficient based on cost alone.

SUGGESTED BUSINESS ACTION

Consider volume, cost and completion rate together, not cost alone.

Q11. How do visit statuses differ across departments and visit types?

BUSINESS QUESTION

How do visit statuses differ across departments and visit types?

BUSINESS OBJECTIVE

Compare completed, cancelled and no-show visits across departments and visit types.

SQL QUERY — PLACEHOLDER

```
SELECT
   dp.DepartmentName,
   v.VisitType,
   v.VisitStatus,
   COUNT(*) AS VisitCount,
   SUM(v.TreatmentCost) AS TreatmentCost
FROM dbo.FactHealthcareVisits v
JOIN dbo.DimDepartment dp
   ON v.DepartmentID = dp.DepartmentID
GROUP BY
   dp.DepartmentName,
   v.VisitType,
   v.VisitStatus;
```

	DepartmentName	VisitType	VisitStatus	VisitCount	TreatmentCost
1	Clinical Department 001	Follow-up	Completed	1	1656.57
2	Clinical Department 013	Routine	Completed	1	8347.81
3	Clinical Department 017	Emergency	Completed	1	3646.40
4	Clinical Department 017	Follow-up	Completed	3	19779.15
5	Clinical Department 025	Follow-up	Completed	1	6929.47
6	Clinical Department 033	Consultation	Completed	1	9593.67
7	Clinical Department 033	Routine	Completed	1	7534.94
8	Clinical Department 037	Consultation	Completed	1	7554.46
9	Clinical Department 045	Follow-up	Completed	1	8567.74
10	Clinical Department 045	Follow-up	No Show	1	0.00
11	Clinical Department 045	Routine	Completed	2	15845.46
12	Clinical Department 053	Emergency	Completed	1	10911.13
13	Clinical Department 057	Routine	Completed	1	5820.73
14	Clinical Department 065	Consultation	Completed	1	650.03
15	Clinical Department 069	Consultation	Cancelled	1	0.00
16	Clinical Department 081	Consultation	Completed	1	8545.81
17	Clinical Department 081	Follow-up	Completed	1	11648.80
18	Clinical Department 085	Follow-up	Completed	1	2778.33

SQL CONCEPTS USED

Explain how your query counts visit statuses by department and visit type.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q11_Visit_Status.csv`

Q11. How do visit statuses differ across departments and visit types?

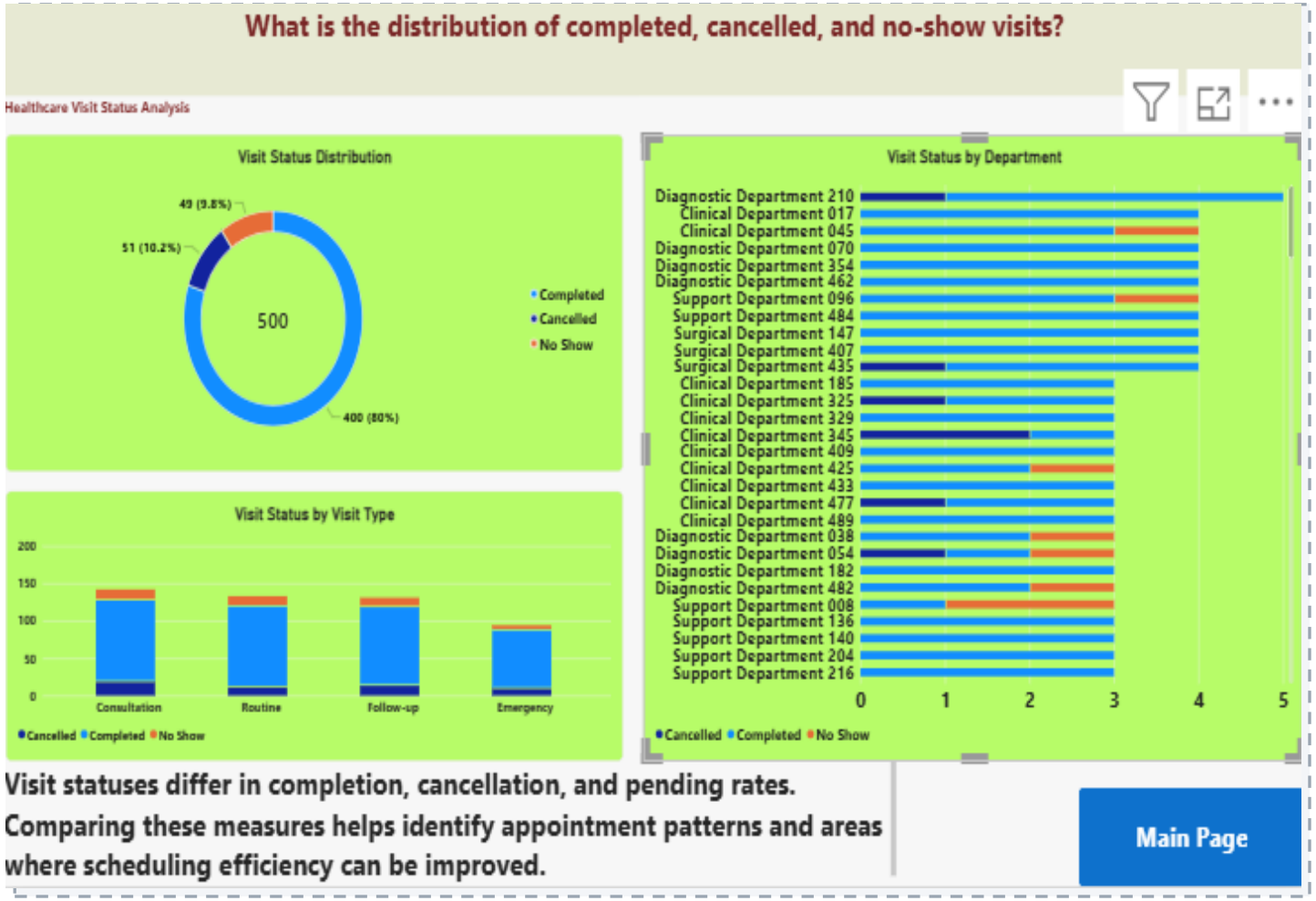


CHART PLACEHOLDERS

1. **Visit Status Distribution**
2. **Visit Status by Department**
3. **Visit Status by Visit Type**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare completed, cancelled, and no-show visits across categories to identify possible opportunities for improving appointment management.

SUGGESTED BUSINESS ACTION

Consider appointment-management ideas only after validating the actual results.

Q12. How can patients be segmented by healthcare utilization?

BUSINESS QUESTION

How can patients be segmented by healthcare utilization?

BUSINESS OBJECTIVE

Group patients by healthcare utilization.

SQL QUERY — PLACEHOLDER

```sql
WITH PatientVisits AS
(
    SELECT
        c.CustomerID,
        c.CustomerName,
        c.Gender,
        c.Region,
        COUNT(v.VisitID) AS VisitFrequency,
        SUM(v.TreatmentCost) AS TotalTreatmentCost,
        AVG(
            CAST(v.PatientAge AS DECIMAL(10,2))
        ) AS AvgPatientAge
    FROM dbo.DimCustomer c
    JOIN dbo.FactHealthcareVisits v
        ON c.CustomerID = v.CustomerID
    GROUP BY
        c.CustomerID,
        c.CustomerName,
        c.Gender,
        c.Region
)
SELECT
    CustomerID,
    CustomerName,
    Gender,
    Region,
    AvgPatientAge,
    VisitFrequency,
    TotalTreatmentCost,
    CASE
        WHEN VisitFrequency >= 5
            THEN 'Frequent Patient'
        WHEN VisitFrequency >= 3
            THEN 'Regular Patient'
        WHEN VisitFrequency = 2
            THEN 'Occasional Patient'
        ELSE 'Single Visit'
    END AS PatientSegment
FROM PatientVisits;
```

	CustomerID	CustomerName	Gender	Region	AvgPatientAge	VisitFrequency	TotalTreatmentCost	PatientSegment
1	1	Vikram Deshmukh 001	Other	West	52.000000	2	2595.77	Occasional Patient
2	2	Aarav Patil 002	Male	West	25.000000	1	2563.79	Single Visit
3	3	Priya Joshi 003	Female	South	58.000000	1	7997.42	Single Visit
4	4	Rohan Joshi 004	Female	West	48.500000	2	14888.93	Occasional Patient
5	5	Aditi Kulkarni 005	Male	West	36.000000	2	7797.03	Occasional Patient
6	6	Aditi Mishra 006	Female	West	52.000000	1	8545.81	Single Visit
7	8	Rahul Pawar 008	Female	West	42.000000	2	15247.44	Occasional Patient
8	9	Aditi Verma 009	Male	West	41.000000	1	0.00	Single Visit
9	11	Karan Mishra 011	Other	West	57.000000	1	1799.35	Single Visit
10	13	Rohan Chavan 013	Female	West	32.000000	1	6470.16	Single Visit
11	15	Aarav Verma 015	Other	South	30.000000	2	19820.75	Occasional Patient
12	16	Arjun Kulkarni 016	Female	West	25.000000	1	0.00	Single Visit
13	17	Rahul Chavan 017	Male	South	59.000000	1	8640.23	Single Visit
14	18	Rahul Chavan 018	Other	West	26.500000	2	11937.73	Occasional Patient

SQL CONCEPTS USED

Explain how your query defines patient segments, only as implemented.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q12_Patient_Segmentation.csv`

Q12. How can patients be segmented by healthcare utilization?

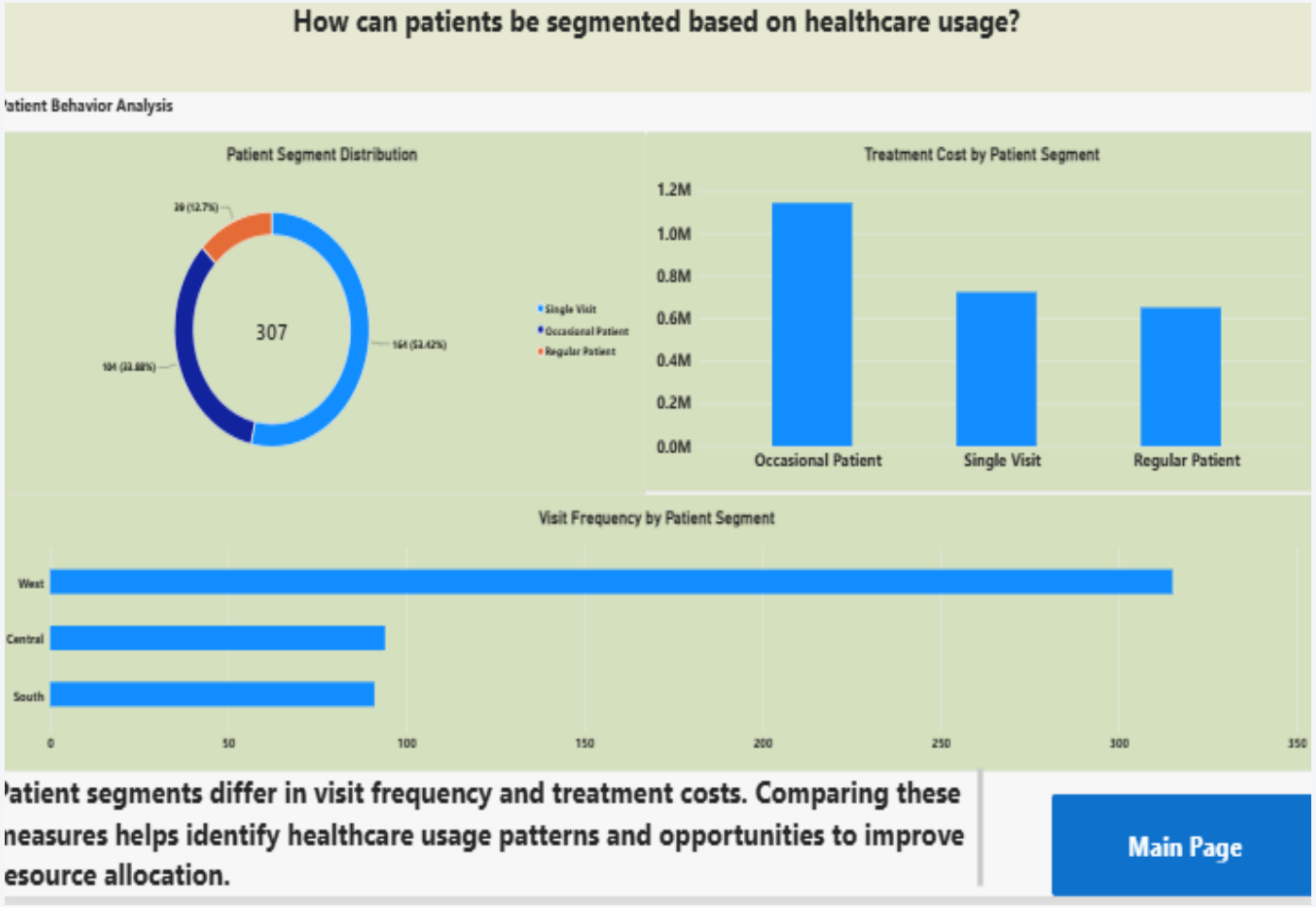


CHART PLACEHOLDERS

1. **Patient Segment Distribution**
2. **Treatment Cost by Patient Segment**
3. **Visit Frequency by Patient Segment**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare visit frequency and treatment costs across the actual patient segments. Do not infer medical conditions or treatment needs that are not represented in the data.

SUGGESTED BUSINESS ACTION

Avoid inferring medical needs that are not represented in the data.

Q13. Which marketing campaigns perform best?

BUSINESS QUESTION

Which marketing campaigns perform best?

BUSINESS OBJECTIVE

Compare campaigns on spend, revenue and return metrics.

SQL QUERY — PLACEHOLDER

```
SELECT
    c.CampaignID,
    c.CampaignName,
    c.CampaignType,
    c.Objective,
    SUM(m.Impressions) AS Impressions,
    SUM(m.Clicks) AS Clicks,
    SUM(m.Conversions) AS Conversions,
    SUM(m.Spend) AS Spend,
    SUM(m.Revenue) AS Revenue,
    CAST(
        100.0 * SUM(m.Clicks)
        / NULLIF(SUM(m.Impressions),0)
        AS DECIMAL(10,2)
    ) AS CTRPct,
    CAST(
        100.0 * SUM(m.Conversions)
        / NULLIF(SUM(m.Clicks),0)
        AS DECIMAL(10,2)
    ) AS ConversionRatePct,
    CAST(
        SUM(m.Revenue)
        / NULLIF(SUM(m.Spend),0)
        AS DECIMAL(10,2)
    ) AS ROAS,
    CAST(
        (SUM(m.Revenue) - SUM(m.Spend))
        / NULLIF(SUM(m.Spend),0) * 100.0
        AS DECIMAL(10,2)
    ) AS ROIPct
FROM dbo.FactMarketing m
JOIN dbo.DimCampaign c
    ON m.CampaignID = c.CampaignID
GROUP BY
    c.CampaignID,
    c.CampaignName,
    c.CampaignType,
    c.Objective;
```

	CampaignID	CampaignName	CampaignType	Objective	Impressions	Clicks	Conversions	Spend	Revenue	CTRPct	ConversionRatePct	ROAS	ROIPct
1	1	Campaign 001	Social	Leads	221463	9636	1092	110086.87	2900515.53	4.35	11.33	26.35	2534.75
2	2	Campaign 002	Search	Conversion	19035	1822	287	3283.79	369698.62	9.57	15.75	112.58	11158.29
3	4	Campaign 004	Social	Awareness	74880	2277	58	18314.09	147701.10	3.04	2.55	8.06	706.49
4	5	Campaign 005	Email	Leads	112582	8417	1620	115953.81	6116034.74	7.48	19.25	52.75	5174.54
5	9	Campaign 009	Social	Awareness	101372	6213	680	14502.67	1378343.00	6.13	10.94	95.04	9404.06
6	10	Campaign 010	Affiliate	Conversion	334903	192...	3136	75581.70	3211341.98	5.75	16.29	42.49	4148.84
7	11	Campaign 011	Email	Retention	184253	105...	959	61895.69	3471505.08	5.71	9.11	56.09	5508.64
8	12	Campaign 012	Affiliate	Leads	121854	5828	951	75956.26	3193382.84	4.78	16.32	42.04	4104.24
9	13	Campaign 013	Display	Conversion	39502	1985	298	64626.72	720436.43	5.03	15.01	11.15	1014.77
10	18	Campaign 018	Social	Leads	40985	4679	729	38120.33	3299089.60	11.42	15.58	86.54	8554.41
11	19	Campaign 019	Email	Leads	164944	114...	2253	84194.12	7942032.48	6.93	19.71	94.33	9333.00

SQL CONCEPTS USED

Explain how your query calculates campaign revenue and return metrics such as ROAS.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q13_Campaign_Performance.csv`

Q13. Which marketing campaigns perform best?

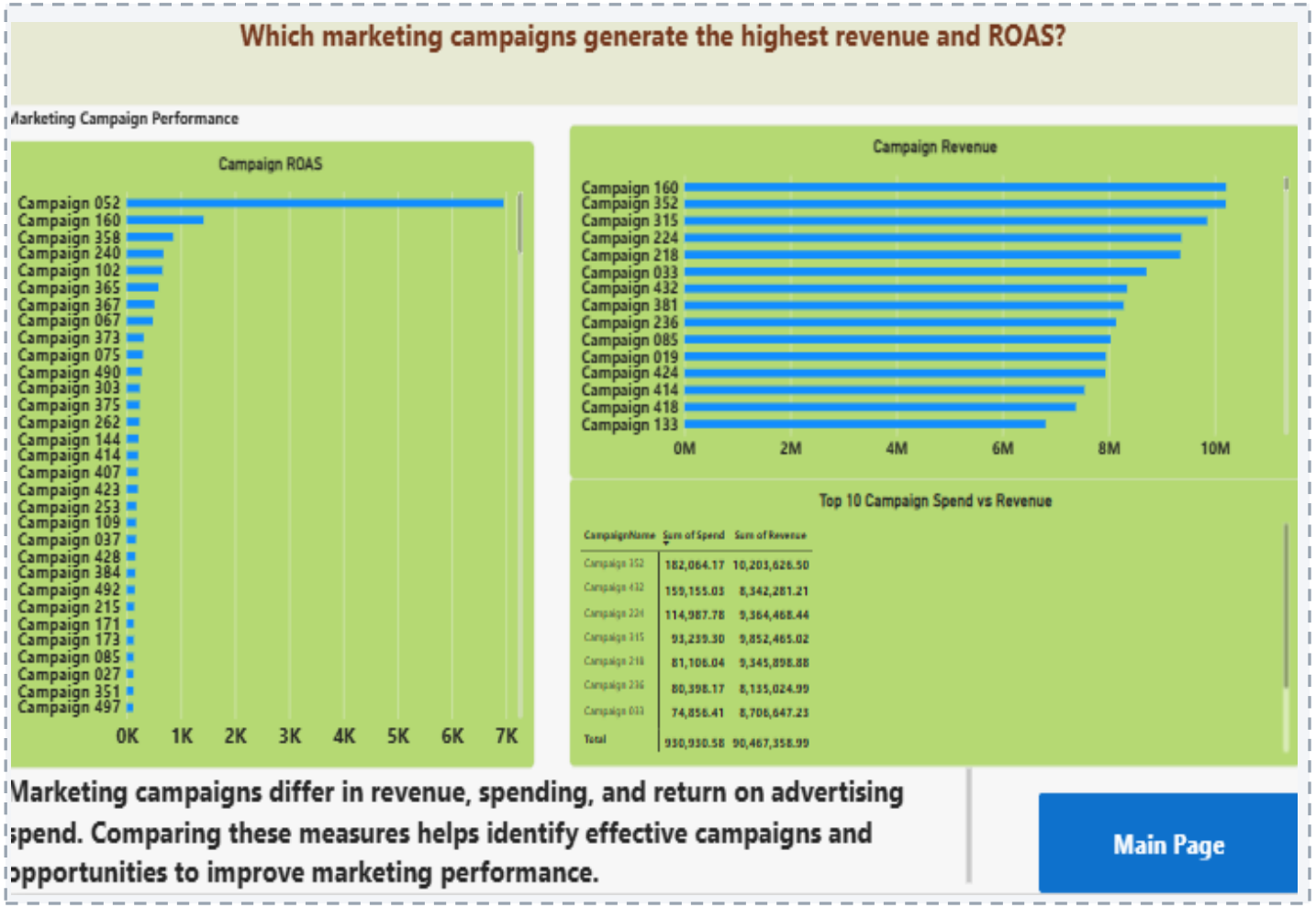


CHART PLACEHOLDERS

1. **Campaign Revenue**
2. **Campaign ROAS**
3. **Campaign Spend vs Revenue**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare campaign spend, revenue, and return metrics using the actual output to identify campaigns worth investigating further.

SUGGESTED BUSINESS ACTION

Validate metric definitions (for example ROAS) before comparing campaigns.

Q14. Which marketing channels deliver the best return?

BUSINESS QUESTION

Which marketing channels deliver the best return?

BUSINESS OBJECTIVE

Compare channels on return and engagement measures.

SQL QUERY — PLACEHOLDER

```
SELECT
  ch.ChannelID,
  ch.ChannelName,
  ch.ChannelType,
  SUM(m.Impressions) AS Impressions,
  SUM(m.Clicks) AS Clicks,
  SUM(m.Conversions) AS Conversions,
  SUM(m.Spend) AS Spend,
  SUM(m.Revenue) AS Revenue,
  CAST(
    100.0 * SUM(m.Clicks)
    / NULLIF(SUM(m.Impressions),0)
    AS DECIMAL(10,2)  ) AS CTRPct, CAST(
    SUM(m.Revenue)
    / NULLIF(SUM(m.Spend),0)  AS DECIMAL(10,2)
  ) AS ROAS,
  CAST( (SUM(m.Revenue) - SUM(m.Spend))
    / NULLIF(SUM(m.Spend),0) * 100.0 AS DECIMAL(10,2) ) AS ROIPct
FROM dbo.FactMarketing m
JOIN dbo.DimChannel ch ON m.ChannelID = ch.ChannelID
GROUP BY
  ch.ChannelID,
  ch.ChannelName,
  ch.ChannelType;
```

	ChannelID	ChannelName	ChannelType	Impressions	Clicks	Conversions	Spend	Revenue	CTRPct	ROAS	ROIP
1	2	Mobile Channel 002	Mobile	97706	3849	881	26372.70	848439.26	3.94	32.17	3117
2	3	Branch Channel 003	Branch	100090	3304	480	47637.81	1696554.25	3.30	35.61	3461
3	5	Social Channel 005	Social	41838	1153	269	17549.82	640564.92	2.76	36.50	3549
4	6	Email Channel 006	Email	10185	741	48	47942.59	144013.61	7.28	3.00	200.3
5	8	Online Channel 008	Online	64061	3880	602	14253.83	309861.05	6.06	21.74	2073
6	12	Social Channel 012	Social	163852	14022	1606	104876.52	2687467.60	8.56	25.63	2462
7	13	Email Channel 013	Email	54605	1019	129	41007.89	471747.90	1.87	11.50	1050
8	14	Partner Channel 014	Partner	18181	575	117	3612.95	306428.36	3.16	84.81	8381
9	16	Mobile Channel 016	Mobile	143680	13209	525	37300.30	657696.97	9.19	17.63	1663
10	17	Branch Channel 017	Branch	15944	666	141	20505.34	402781.02	4.18	19.64	1864
11	18	ATM Channel 018	ATM	43120	792	153	4028.98	376661.77	1.84	93.49	9248
12	19	Social Channel 019	Social	99026	8498	947	83064.33	1388128.68	8.58	16.71	1571
13	20	Email Channel 020	Email	52807	6127	351	41299.91	728456.53	11.60	17.64	1663

SQL CONCEPTS USED

Explain how your query calculates return and engagement measures by channel.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q14_Channel_ROI.csv`

Q14. Which marketing channels deliver the best return?

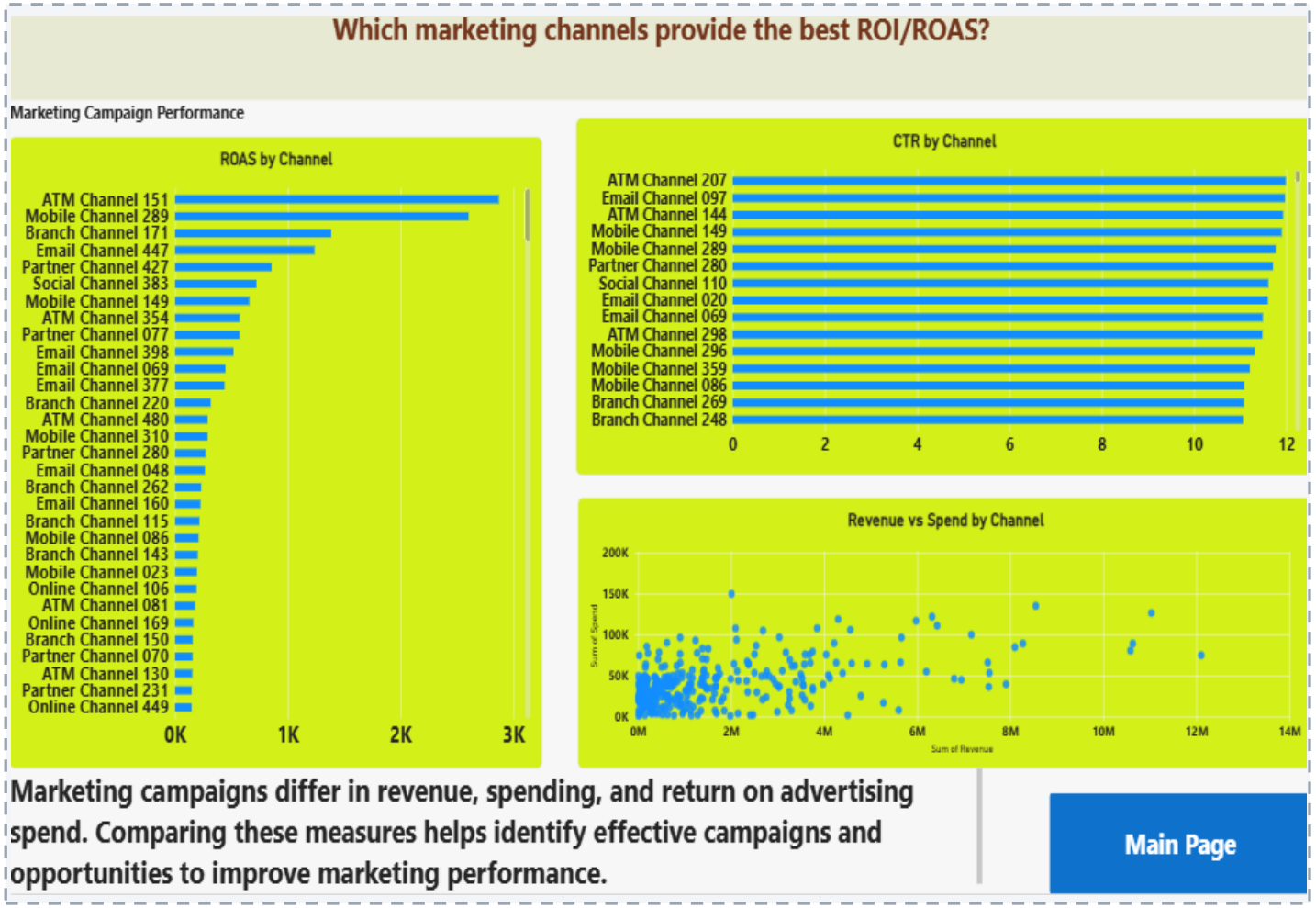


CHART PLACEHOLDERS

1. **ROAS by Channel**
2. **Revenue vs Spend by Channel**
3. **CTR by Channel**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Channel performance may vary across return and engagement measures. Recommend reallocating budgets only after validating the actual results and calculation definitions.

SUGGESTED BUSINESS ACTION

Validate calculation definitions before any budget reallocation.

Q15. Which channels acquire the most customers over time?

BUSINESS QUESTION

Which channels acquire the most customers over time?

BUSINESS OBJECTIVE

Compare customer acquisition by channel and time period.

SQL QUERY — PLACEHOLDER

```sql
WITH AcquisitionTouch AS
(
    SELECT
        m.CustomerID,
        m.ChannelID,
        m.CampaignID,
        d.FullDate AS TouchDate,
        ROW_NUMBER() OVER
        (
            PARTITION BY m.CustomerID
            ORDER BY d.FullDate DESC, m.MarketingID DESC
        ) AS rn
    FROM dbo.FactMarketing m
    JOIN dbo.DimDate d
        ON m.DateID = d.DateID
    JOIN dbo.DimCustomer c
        ON m.CustomerID = c.CustomerID
    WHERE d.FullDate <= c.SignupDate
      AND d.FullDate >= DATEADD(DAY,-30,c.SignupDate)
),
Acquisition AS
(
    SELECT
        c.CustomerID,
        c.SignupDate,
        CASE
            WHEN a.CustomerID IS NULL
                THEN 'Organic'
            ELSE ch.ChannelName
        END AS AcquisitionChannel,
        CASE
            WHEN a.CustomerID IS NULL
                THEN NULL
            ELSE cp.CampaignName
        END AS AcquisitionCampaign
    FROM dbo.DimCustomer c
    LEFT JOIN AcquisitionTouch a
        ON c.CustomerID = a.CustomerID
       AND a.rn = 1
    LEFT JOIN dbo.DimChannel ch
        ON a.ChannelID = ch.ChannelID
    LEFT JOIN dbo.DimCampaign cp
        ON a.CampaignID = cp.CampaignID
)
SELECT
    AcquisitionChannel,
    YEAR(SignupDate) AS SignupYear,
    MONTH(SignupDate) AS SignupMonth,
    COUNT(*) AS AcquiredCustomers
FROM Acquisition
GROUP BY
    AcquisitionChannel,
    YEAR(SignupDate),
    MONTH(SignupDate);
```

	AcquisitionChannel	SignupYear	SignupMonth	AcquiredCustomers
1	ATM Channel 151	2024	8	1
2	Branch Channel 108	2024	5	1
3	Branch Channel 311	2024	3	1
4	Mobile Channel 205	2025	4	1
5	Mobile Channel 282	2025	5	1
6	Mobile Channel 471	2024	7	1
7	Organic	2024	1	25
8	Organic	2024	2	31
9	Organic	2024	3	31
10	Organic	2024	4	34
11	Organic	2024	5	39
12	Organic	2024	6	28
13	Organic	2024	7	30
14	Organic	2024	8	27
15	Organic	2024	9	27
16	Organic	2024	10	36
17	Organic	2024	11	25
18	Organic	2024	12	36

SQL CONCEPTS USED

Explain how your query counts acquired customers by channel and period.

List only concepts that appear in your actual query.

OUTPUT FILE `SQL_Outputs/Q15_Customer_Acquisition.csv`

Q15. Which channels acquire the most customers over time?

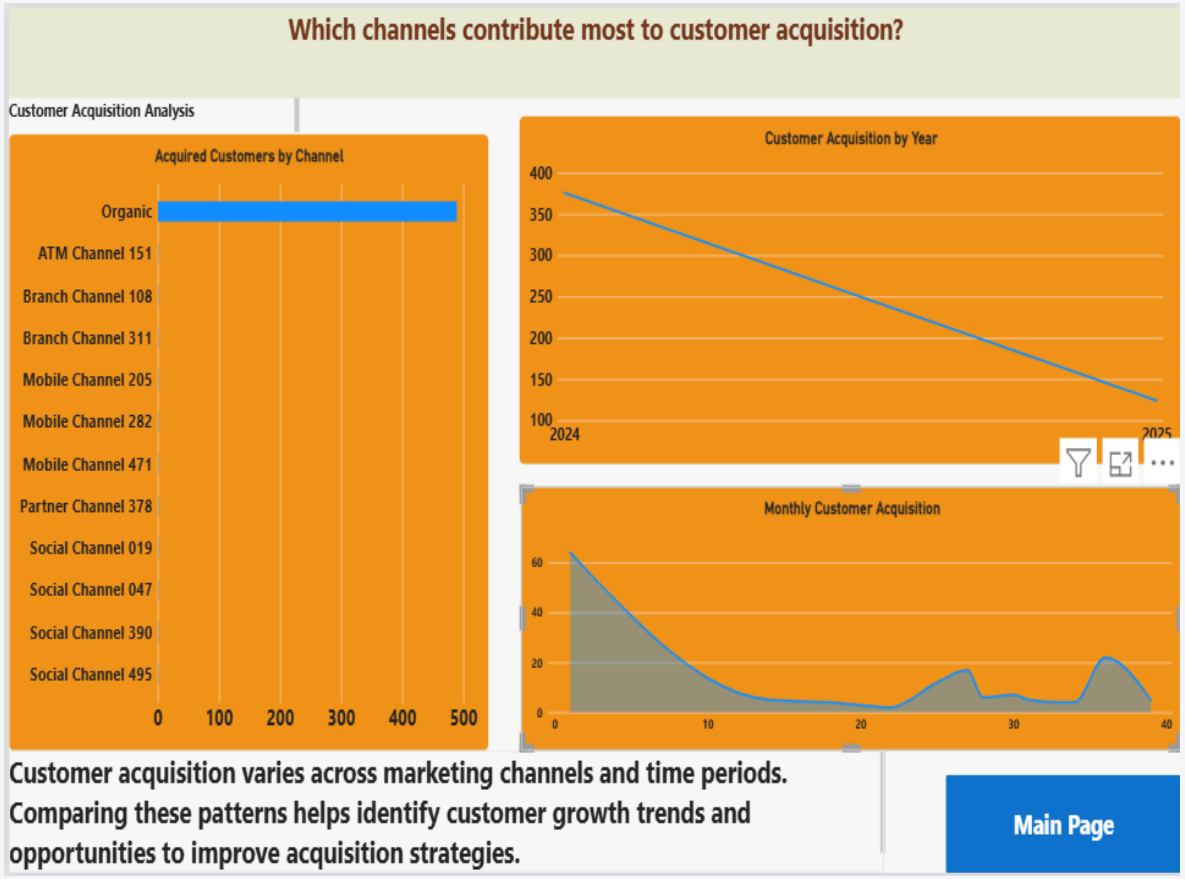


CHART PLACEHOLDERS

1. **Acquired Customers by Channel**
2. **Monthly Customer Acquisition**
3. **Customer Acquisition by Year**

PROVISIONAL INSIGHT (EXAMPLE – VALIDATE)

Compare customer acquisition by channel and time period to identify observed acquisition patterns. Do not claim that a channel is cost-effective without acquisition cost data.

SUGGESTED BUSINESS ACTION

Avoid cost-effectiveness claims without acquisition cost data.

Key Learnings



SQL analysis

Writing queries to answer defined business questions.



Relational data

Working with related tables, keys and date fields.



Dashboard design

Turning query outputs into clear Power BI pages.



Communicating insights

Explaining findings and limits to business audiences.

Business Value

Retail and E-commerce	Review sales, profit, product, customer and regional performance.
BFSI	Review loan approvals, default risk, transactions and customer value.
Healthcare	Review visits, departments, visit status and patient utilization.
Marketing	Review campaign performance, channel return and acquisition.

These are potential decision-support uses. Business value depends on validated results; no unverified financial improvement is claimed.

Thank You



Aman Ade

SQL Server | Power BI | CSV | GitHub